Series Estimation of Cointegrated System with Time Varying Coefficients*

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1 — Introduction

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2.1. Model setup

We consider the time-varying coefficients cointegrated system (cf. Phillips 1991)

$$\mathbf{y}_t = \mathbf{A}_t \mathbf{X}_t + \mathbf{u}_{0t}, \tag{1}$$

$$\mathbf{X}_t = \mathbf{X}_{t-1} + \mathbf{u}_{xt}, \tag{2}$$

where $t = 1, 2, \cdots, n$. $\mathbf{y}_t$ denotes a d-dimensional vector, and $\mathbf{X}_t$ is a $K_1 \times 1$ vector of integrated regressors. The matrix of time-varying cointegrating coefficients is given by $\mathbf{A}_t := \mathbf{A}(\tau_t)$, where $\mathbf{A}_t$ is $d \times K_1$ with elements modeled as functions of scaled time point $\tau_t := t/n$. The process $(\mathbf{X}_t)_t$ is initialized at $\mathbf{X}_0 = O_p(1)$.

The innovations $\mathbf{u}_t = \left(\mathbf{u}_{0t}^\top, \mathbf{u}_{xt}^\top\right)^\top$ are determined according to the linear process

$$\mathbf{u}_t = \boldsymbol{\Phi}(\mathcal{L})\boldsymbol{\varepsilon}_t = \sum_{j=0}^{\infty} \boldsymbol{\Phi}_j \boldsymbol{\varepsilon}_{t-j}, \tag{3}$$

where $\boldsymbol{\Phi}(\mathcal{L}) = \sum_{j=0}^{\infty} \boldsymbol{\Phi}_j \mathcal{L}^j$, $(\boldsymbol{\Phi}_j)_j$ is a sequence of $(d+K_1) \times (d+K_1)$ matrices, $\mathcal{L}$ is the lag operator, and $(\boldsymbol{\varepsilon}_t)_t$ is a sequence of independent and identically distributed ($i.i.d.$) random vectors with dimension $d + K_1$. We impose the following conditions.

Assumption 1. (i) $(\boldsymbol{\varepsilon}_t)_t$ is an $i.i.d.$ $(d+K_1)$-dimensional random vector with $\mathbb{E}(\boldsymbol{\varepsilon}_t) = \mathbf{0}_{(d+K_1)\times 1}$, $\boldsymbol{\Sigma}_\varepsilon := \mathbb{E}(\boldsymbol{\varepsilon}_t \boldsymbol{\varepsilon}_t^\top)$, where $\lambda_{\min}(\boldsymbol{\Sigma}_\varepsilon) > 0$, and $\mathbb{E}\|\boldsymbol{\varepsilon}_t\|^p < \infty$, for some $p > 4$. (ii) The coefficient matrices in (3) satisfy $\sum_{j=0}^{\infty} j\|\boldsymbol{\Phi}_j\| < \infty$ and $|\boldsymbol{\Phi}(1)| \neq 0$.

Under Assumption 1, $\mathbf{u}_t$ has covariance matrix $\boldsymbol{\Sigma} = \sum_{j=0}^{\infty} \boldsymbol{\Phi}_j \boldsymbol{\Sigma}_\varepsilon \boldsymbol{\Phi}_j^\top$, with $\mathbb{E}\|\mathbf{u}_t\|^p < \infty$, for some $p > 4$. By functional central limit theory (FCLT) for linear process (cf. Phillips and Solo 1992), we have

$$\frac{1}{\sqrt{n}} \sum_{t=1}^{\lfloor nr \rfloor} \mathbf{u}_t \Rightarrow \mathbf{B}(r) \ \text{ on } \ D\big([0,1],\, \mathbb{R}^{d+K_1}\big), \tag{4}$$

where $\mathbf{B}(r)$ denotes a vector Brownian motion with long-run covariance matrix $\boldsymbol{\Omega} = \boldsymbol{\Phi}(1)\boldsymbol{\Sigma}_\varepsilon\boldsymbol{\Phi}(1)^\top >$

0 and $\mathbf{\Phi}(1) = \sum_{j=0}^{\infty} \mathbf{\Phi}_j$. For the ease of theoretical development, we partition $\mathbf{\Sigma}$, $\mathbf{\Phi}(1)$, and $\mathbf{\Omega}$ comfortably with $\mathbf{u}_t$ as

$$\mathbf{\Sigma} = \begin{bmatrix} \mathbf{\Sigma}_{00} & \mathbf{\Sigma}_{0x} \\ \mathbf{\Sigma}_{x0} & \mathbf{\Sigma}_{xx} \end{bmatrix}, \quad \mathbf{\Phi}(1) = \begin{bmatrix} \mathbf{\Phi}_0(1) \\ \mathbf{\Phi}_x(1) \end{bmatrix}, \quad \mathbf{\Omega} = \begin{bmatrix} \mathbf{\Omega}_{00} & \mathbf{\Omega}_{0x} \\ \mathbf{\Omega}_{x0} & \mathbf{\Omega}_{xx} \end{bmatrix}, \tag{5}$$

where $\mathbf{\Phi}_0(1)$ and $\mathbf{\Phi}_x(1)$ are of dimension $d \times (d + K_1)$ and $K_1 \times (d + K_1)$, respectively. Using the Beveridge–Nelson (BN) decomposition, we have the following decomposition for $\mathbf{u}_t$:

$$\mathbf{u}_t = \mathbf{\Phi}(1)\boldsymbol{\varepsilon}_t - \Delta\tilde{\boldsymbol{u}}_t, \ \text{ for } \ \tilde{\mathbf{u}}_t = \sum_{j=0}^{\infty} \widetilde{\mathbf{\Phi}}_j \boldsymbol{\varepsilon}_{t-j}, \quad \widetilde{\mathbf{\Phi}}_j = \sum_{k=j+1}^{\infty} \mathbf{\Phi}_j, \tag{6}$$

where Assumption 1(ii) ensures $\sum_{j=0}^{\infty} \left\| \widetilde{\mathbf{\Phi}}_j \right\| < \infty$ (Phillips and Solo 1992). Corresponding to the partition of $\mathbf{u}_t$, we write

$$\mathbf{u}_{0t} = \sum_{j=0}^{\infty} \mathbf{\Phi}_{0j}\boldsymbol{\varepsilon}_{t-j}, \quad \tilde{\mathbf{u}}_{0t} = \sum_{j=0}^{\infty} \widetilde{\mathbf{\Phi}}_{0j}\boldsymbol{\varepsilon}_{t-j}, \tag{7}$$

and also

$$\mathbf{u}_{xt} = \sum_{j=0}^{\infty} \mathbf{\Phi}_{xj}\boldsymbol{\varepsilon}_{t-j}, \quad \tilde{\mathbf{u}}_{xt} = \sum_{j=0}^{\infty} \tilde{\mathbf{\Phi}}_{xj}\boldsymbol{\varepsilon}_{t-j}, \tag{8}$$

where $\mathbf{\Phi}_{0j}$, $\widetilde{\mathbf{\Phi}}_{0j}$ are of dimension $d \times (d + K_1)$ and $\mathbf{\Phi}_{xj}$, $\widetilde{\mathbf{\Phi}}_{xj}$ are of dimension $K_1 \times (d + K_1)$, respectively. In addition, the limit theory also involves the partitioned components of the one-sided long-run covariance matrix

$$\mathbf{\Lambda} := \begin{bmatrix} \mathbf{\Lambda}_{00} & \mathbf{\Lambda}_{0x} \\ \mathbf{\Lambda}_{x0} & \mathbf{\Lambda}_{xx} \end{bmatrix} = \sum_{j=1}^{\infty} \mathbb{E}\left(\mathbf{u}_t \mathbf{u}_{t-j}^{\top}\right).$$

As usual, we have $\mathbf{\Omega} = \mathbf{\Sigma} + \mathbf{\Lambda} + \mathbf{\Lambda}^{\top}$ and

$$\mathbf{\Lambda} = \sum_{j=1}^{\infty}\sum_{k=0}^{\infty} \mathbf{\Phi}_{k+j}\mathbf{\Sigma}_{\varepsilon}\mathbf{\Phi}_k^{\top} = \sum_{k=0}^{\infty}\left(\sum_{j=1}^{\infty} \mathbf{\Phi}_{k+j}\right)\mathbf{\Sigma}_{\varepsilon}\mathbf{\Phi}_k^{\top} = \sum_{k=0}^{\infty} \widetilde{\mathbf{\Phi}}_k \mathbf{\Sigma}_{\varepsilon}\mathbf{\Phi}_k^{\top} = \mathbb{E}\left(\tilde{\mathbf{u}}_t \mathbf{u}_t^{\top}\right).$$

2.2. Estimation procedure

Our objective is to develop estimators for the time-varying cointegrating coefficients $\mathbf{A}(\tau_t)$ and the time-invariant coefficients $\mathbf{\Gamma}$. We first state the assumption imposed on $\mathbf{A}(\tau_t)$:

Assumption 2. Let $\mathbf{A}^{(i,j)}(\tau_t)$ be the (i,j)th elements in $\mathbf{A}(\tau_t)$, where $i = 1, 2, \cdots, d$, $j = 1, 2, \cdots, K$. $\mathbf{A}^{(i,j)}(\cdot)$ is a squared integrable real function: $\mathbf{A}^{(i,j)}(\cdot) \in L^2(0,1) = \left\{ a(r) : \int_0^1 a^2(r)dr < \infty \right\}$ and is $q+1$ times continuously differentiable on $[0,1]$, where $q > 3$.

Under Assumption 2, $\mathbf{A}(\tau_t)$ admits an orthogonal expansion

$$\mathbf{A}(\tau_t) = \sum_{i=0}^{\infty} \mathbf{B}_i \psi_i(\tau_t),$$

where $\mathbf{B}_i$ is of dimension $d \times K_1$ and

$$\psi_0(\tau_t) = 1, \quad \psi_j(\tau_t) = \sqrt{2}\cos\left(j\pi\tau_t\right), \tag{9}$$

where $j = 1, 2, 3, \cdots$. Then, $\{\psi_j(r)\}_j$ is an orthonormal basis in $L^2[0,1]$. (9) defines the Chebyshev time polynomials, first used in econometrics by Bierens 1997 in the context of testing a unit root against a nonlinear trend–stationary alternative. They are also employed by Dong and Linton 2018 for estimating deterministic time trends in additive nonparametric models. Some basic properties of the Chebyshev time polynomials are summarized in Section D.

Let k be a positive integer. We then define a truncated series

$$\mathbf{A}_k(\tau_t) = \sum_{i=0}^{k-1} \mathbf{B}_i^{(k)} \psi_i(\tau_t) = \mathbf{B}^{(k)} \left(\boldsymbol{\psi}_{(k)}(\tau_t) \otimes \mathbf{I}_{K_1} \right), \tag{10}$$

where $\boldsymbol{\psi}_{(k)}(\tau_t) = \left(\psi_0(\tau_t), \psi_1(\tau_t), \cdots, \psi_{k-1}(\tau_t)\right)^\top$. Under Assumption 2, by directly applying Theorem D.1 in Appendix D, we have

$$\|\mathbf{A}(\tau_t) - \mathbf{A}_k(\tau_t)\| = O\left(\frac{1}{k^q}\right), \tag{11}$$

for any $\tau_t \in [0,1]$. This shows that $\mathbf{A}(\tau_t)$ can be well approximated by (10), and the approximation error goes to zero as $k \to \infty$.

By replacing $\mathbf{A}(\tau_t)$ in (1) with (10), we have the approximated model:

$$\mathbf{y}_t = \sum_{i=0}^{k-1} \mathbf{B}_i \psi_i(\tau_t) \mathbf{X}_t + \mathbf{u}_{0t}^{(k)}, \tag{12}$$

where $\mathbf{u}_{0t}^{(k)} = \mathbf{u}_{0t} + (\mathbf{A}_t - \mathbf{A}_k(\tau_t))\, \mathbf{X}_t$. Define

$$\begin{aligned} \underset{kK_1 \times 1}{\mathbf{X}_t^{(k)}} &= \left(\mathbf{X}_t^\top, \psi_1(\tau_t)\mathbf{X}_t^\top, \cdots, \psi_{k-1}(\tau_t)\mathbf{X}_t^\top\right)^\top \\ \underset{d \times kK_1}{\mathbf{B}^{(k)}} &= (\mathbf{B}_0, \mathbf{B}_1, \cdots, \mathbf{B}_{k-1}). \end{aligned} \tag{13}$$

We can write (12) more conveniently as

$$\mathbf{y}_t = \mathbf{B}^{(k)} \mathbf{X}_t^{(k)} + \mathbf{u}_{0t}^{(k)}. \tag{14}$$

$\mathbf{B}^{(k)}$ can thus be estimated by LS (least squares):

$$\widehat{\mathbf{B}}_n^{(k)} = \left(\sum_{t=1}^n \mathbf{y}_t \mathbf{X}_t^{(k)\top}\right) \left(\sum_{t=1}^n \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top}\right)^{-1}$$

Then, for $\tau_t \in [0,1]$, the series estimator of $\mathbf{A}(\tau_t)$ is

$$\widehat{\mathbf{A}}_n^{(k)}(\tau_t) = \widehat{\mathbf{B}}_n^{(k)} \mathbf{T}_{(k)}(\tau_t), \tag{15}$$

based on (10), where $\mathbf{T}_{(k)}(\tau_t) = \boldsymbol{\psi}_{(k)}(\tau_t) \otimes \mathbf{I}_{K_1}$.

2.3. Asymptotic theory

We now introduce an additional assumption on the series expansion order k. As expected, it is inversely related to q: the smoother the function $\mathbf{A}(\tau_t)$, the lower the required order of the series expansion.

Assumption 3. $k = \lfloor c \cdot n^\kappa \rfloor$ is a sequence of integers for some $c > 0$, where $\frac{1}{q} < \kappa < \min\left\{\frac{1}{3}, \frac{1}{2} - \frac{1}{p}\right\}$.

Theorem 1 derives the first-order asymptotic expansion of the series estimator (15) and establishes its leading term.

Theorem 1. Suppose that Assumptions 1-3 are satisfied. For any given $\tau_t \in [0,1]$, we have, as $n \to \infty$

$$n \cdot \left(\widehat{\mathbf{A}}_n^{(k)}(\tau_t) - \mathbf{A}(\tau_t) + \mathbf{\Gamma}(\tau_t)\right) = \mathbf{\Gamma}_{ux}^{(k)} \left(\mathbf{\Lambda}_{xx}^{(k)}\right)^{-1} \mathbf{T}_{(k)}(\tau_t) + o_p(\sqrt{k}), \tag{16}$$

where $\mathbf{\Gamma}(\tau_t) := \mathbf{A}(\tau_t) - \mathbf{A}_k(\tau_t)$ and

$$\mathbf{\Lambda}_{xx}^{(k)} := \int_0^1 \left\{\boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r) \otimes \left(\mathbf{B}_X(r)\mathbf{B}_X^\top(r)\right)\right\} dr,$$
$$\mathbf{\Gamma}_{ux}^{(k)} := \int_0^1 d\mathbf{B}_0(r) \left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r)\right)^\top + \boldsymbol{\ell}_{1(k)} \otimes \left(\mathbf{\Lambda}_{0x} + \mathbf{\Sigma}_{0x}\right).$$

Note that (16) is only a stochastic expansion and does not by itself imply a mixed normal limit, since $\mathbf{B}_X(r)$ and $\mathbf{B}_0(r)$ are correlated. Define

$$\widehat{\mathbb{V}}_{n(k)}(\tau_t) = \left\{\mathbf{T}_{(k)}^\top(\tau_t)\mathbf{\Lambda}_{n,xx}^{-1}\mathbf{T}_{(k)}(\tau_t)\right\} \otimes \widehat{\mathbf{\Omega}}_{n,00},$$

where $\widehat{\mathbf{\Omega}}_{n,00}$ is a consistent estimator of $\mathbf{\Omega}_{00}$. In the next corollary, we show that, mixed Normal asymptotic distribution is achieved in the pure cointegration case when $\mathbf{X}_t$ is strictly exogenous.

Corollary 1. Suppose that conditions in Theorem 1 are satisfied and $\mathbb{E}\left(\mathbf{u}_{0t} \otimes \mathbf{u}_{xs}\right) = \mathbf{0}_{dK_1 \times 1}$ for any t, s, we have

$$n \left(\widehat{\mathbb{V}}_{n(k)}(\tau_t)\right)^{-1/2} \mathrm{vec}\left(\widehat{\mathbf{A}}_n^{(k)}(\tau_t) - \mathbf{A}(\tau_t) + \mathbf{\Gamma}(\tau_t)\right) \xrightarrow{d} \mathcal{N}\left(\mathbf{0}_{dK_1 \times 1}, \mathbf{I}_{dK_1}\right), \tag{17}$$

for any fixed $\tau_t \in [0,1]$.

In view of the fact that $\left\|\widehat{\mathbb{V}}_{n(k)}(\tau_t)\right\| = O(1)k$, the convergence rate in (17) is $O_p(1)\frac{n}{\sqrt{k}}$. The convergence rate is comparable with the kernel estimate in the literature. Theorem 3.1 in Phillips, Li, and Gao 2017 shows that the *type I* super convergence rate is given by $nh^{1/2}$, where h is the bandwidth parameter. Thinking of k^{-1} as equivalent to the bandwidth h, the convergence rate in the two situations are quite comparable. However, due to kernel estimation degeneracy issues and rotation matrix introduced in their approach, the kernel estimate also involves a slower *type II* convergence rate, which is given by nh.

3 — Including stationary regressors

In this section, we consider a refinement of the model (1) in which stationary regressors are added to model the "short run" effects:

$$\mathbf{y}_t = \mathbf{A}_t \mathbf{X}_t + \mathbf{\Gamma} \mathbf{Z}_t + \mathbf{u}_{0t}, \tag{18}$$

where $\mathbf{Z}_t$ is a K_2-dimensional vector of stationary variables. It is worth mentioning that $\mathbf{Z}_t$ enter the system (18)-(2) with time-invariant coefficients $\mathbf{\Gamma}$. Allowing these coefficients to vary over time would undermine the interpretation of $\mathbf{A}_t$ as the evolving long-run relation between $\mathbf{y}_t$ and $\mathbf{X}_t$, since a time-varying $\mathbf{\Gamma}_t$ interacting with stationary regressors would introduce additional low-frequency components in $\mathbf{y}_t$ that are incompatible with the cointegration structure. Fixing $\mathbf{\Gamma}$ ensures that all systematic time variation in the long-run component is captured solely by $\mathbf{A}_t$, while $\mathbf{Z}_t$ accounts only for transitory fluctuations. This separation guarantees that the contribution of stationary regressors remains purely short-run, without contaminating the low-frequency variation driven by the integrated process $\mathbf{X}_t$ and prevents spurious drift that would arise from time-varying coefficients on $\mathbf{Z}_t$.

We assume that $\left(\mathbf{u}_t^\top, \mathbf{Z}_t^\top\right)^\top$ satisfies the following invariance principle:

$$\frac{1}{\sqrt{n}} \sum_{t=1}^{\lfloor nr \rfloor} \left(\mathbf{u}_t^\top, \mathbf{Z}_t^\top\right)^\top \Rightarrow \mathbf{B}(r) \ \text{ on } \ D\big([0,1],\, \mathbb{R}^{d+K_1+K_2}\big)\,, \tag{19}$$

where $\mathbf{B}(r) = \left(\mathbf{B}_0^\top(r), \mathbf{B}_X^\top(r), \mathbf{B}_Z^\top(r)\right)^\top$ is a vector Brownian motion with long-run covariance matrix $\mathbf{\Omega}$. We have $\mathbf{\Omega} = \mathbf{\Sigma} + \mathbf{\Lambda} + \mathbf{\Lambda}^\top$ and the following three-way partition

$$\mathbf{\Omega} := \begin{bmatrix} \mathbf{\Omega}_{00} & \mathbf{\Omega}_{0x} & \mathbf{\Omega}_{0z} \\ \mathbf{\Omega}_{x0} & \mathbf{\Omega}_{xx} & \mathbf{\Omega}_{xz} \\ \mathbf{\Omega}_{z0} & \mathbf{\Omega}_{zx} & \mathbf{\Omega}_{zz} \end{bmatrix}, \quad \mathbf{\Lambda} := \begin{bmatrix} \mathbf{\Lambda}_{00} & \mathbf{\Lambda}_{0x} & \mathbf{\Lambda}_{0z} \\ \mathbf{\Lambda}_{x0} & \mathbf{\Lambda}_{xx} & \mathbf{\Lambda}_{xz} \\ \mathbf{\Lambda}_{z0} & \mathbf{\Lambda}_{zx} & \mathbf{\Lambda}_{zz} \end{bmatrix}.$$

With the estimation procedure introduced in Section 2.2, we now have an extended model:

$$\mathbf{y}_t = \mathbf{B}^{(k)}\mathbf{X}_t^{(k)} + \mathbf{\Gamma}\mathbf{Z}_t + \mathbf{u}_{0t}^{(k)}, \tag{20}$$

where $\mathbf{u}_{0t}^{(k)} = \mathbf{u}_{0t} + (\mathbf{A}_t - \mathbf{A}_k(\tau_t))\,\mathbf{X}_t$ and $\mathbf{B}^{(k)}$, $\mathbf{X}_t^{(k)}$ are defined as in (13). $\mathbf{B}^{(k)}$ and $\mathbf{\Gamma}$ can thus be estimated by LS (least squares):

$$\begin{bmatrix} \widehat{\mathbf{B}}_n^{(k)} & \widehat{\mathbf{\Gamma}}_n \end{bmatrix} = \begin{bmatrix} \sum_{t=1}^n \mathbf{y}_t\mathbf{X}_t^{(k)\top} \\ \sum_{t=1}^n \mathbf{y}_t\mathbf{Z}_t^\top \end{bmatrix} \begin{bmatrix} \sum_{t=1}^n \mathbf{X}_t^{(k)}\mathbf{X}_t^{(k)\top} & \sum_{t=1}^n \mathbf{X}_t^{(k)}\mathbf{Z}_t^\top \\ \sum_{t=1}^n \mathbf{Z}_t\mathbf{X}_t^{(k)\top} & \sum_{t=1}^n \mathbf{Z}_t\mathbf{Z}_t^\top \end{bmatrix}^{-1}.$$

Finally, the series estimator for $\mathbf{A}(\cdot)$ is obtained in the same manner as in (15).

Our next theorem characterises the limit theory of the estimator for $\mathbf{\Gamma}$, the coefficient matrix of the stationary regressors.

Theorem 2. Suppose that Assumptions 1-3 are satisfied. If, in addition, (19) holds and $\mathbb{E}\left(\mathbf{u}_{0t} \otimes \mathbf{Z}_t\right) = \mathbf{0}_{dK_2\times 1}$, then

$$\sqrt{n}\cdot \text{vec}\left(\widehat{\mathbf{\Gamma}}_n - \mathbf{\Gamma}\right) \xrightarrow{d} \mathcal{N}\left(\mathbf{0}_{dK_2\times 1}, \left(\mathbf{\Sigma}_{zz}^{-1}\otimes \mathbf{I}_d\right)\mathbf{\Omega}^{(0z)}\left(\mathbf{\Sigma}_{zz}^{-1}\otimes \mathbf{I}_d\right)\right),$$

where $\mathbf{\Omega}^{(0z)} = \sum_{j=-\infty}^{\infty}\mathbb{E}\left(\mathbf{u}_{0t}\mathbf{u}_{0t+j}^\top \otimes \mathbf{Z}_t\mathbf{Z}_{t+j}^\top\right)$.

Of course, we need the condition $\mathbb{E}\left(\mathbf{u}_{0t}\otimes \mathbf{Z}_t\right) = \mathbf{0}_{dK_2\times 1}$, so that $\mathbf{Z}_t$ is weakly exogenous and $\mathbf{\Gamma}$ can be consistently estimated. Interestingly, the asymptotic distribution of $\widehat{\mathbf{\Gamma}}_n$ coincides with the case when the cointegrating coefficients $\mathbf{A}_t$ are constant (e.g., Park and Phillips 1989, Phillips 1995), provided that series expansion order k satisfies Assumption 3. The reason is that, when k grows sufficiently fast, the series approximation error becomes asymptotically negligible, so that it does not contribute to the first order asymptotic expansion of $\sqrt{n}\cdot\text{vec}\left(\widehat{\mathbf{\Gamma}}_n - \mathbf{\Gamma}\right)$, and the standard $\sqrt{n}$–asymptotic theory for regressions with stationary regressors remains valid.

Theorem 3 below establishes the first-order asymptotic expansion for the series estimator (15) for the time-varying coefficients $\mathbf{A}(\cdot)$.

Theorem 3. Suppose that Assumptions 1-3 are satisfied. If, in addition, (19) holds. For any given

$\tau_t \in [0,1]$, we have, as $n \to \infty$

$$n \cdot \left(\widehat{\mathbf{A}}_n^{(k)}(\tau_t) - \mathbf{A}(\tau_t) + \mathbf{\Gamma}(\tau_t) \right) = \widetilde{\mathbf{\Gamma}}_{ux}^{(k)} \left(\mathbf{\Lambda}_{xx}^{(k)} \right)^{-1} \mathbf{T}_{(k)}(\tau_t) + o_p\left(\sqrt{k}\right), \tag{21}$$

where $\mathbf{\Gamma}(u) := \mathbf{A}(u) - \mathbf{A}_k(u)$, $\mathbf{\Lambda}_{xx}^{(k)}$ is defined as in Theorem 1, and

$$\begin{aligned}\widetilde{\mathbf{\Gamma}}_{ux}^{(k)} := \int_0^1 \left(d\mathbf{B}_0(r) - \mathbf{\Sigma}_{0z}\mathbf{\Sigma}_{zz}^{-1} d\mathbf{B}_Z(r) \right) \left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r) \right)^\top + \boldsymbol{\ell}_{1(k)} \otimes (\mathbf{\Lambda}_{0x} + \mathbf{\Sigma}_{0x}) \\ - \boldsymbol{\ell}_{1(k)} \otimes \left\{ \mathbf{\Sigma}_{0z}\mathbf{\Sigma}_{zz}^{-1} (\mathbf{\Lambda}_{zx} + \mathbf{\Sigma}_{zx}) \right\}.\end{aligned}$$

Compared to Theorem 1, we see that the leading term is altered once stationary regressors are included, unless $\mathbb{E}(\mathbf{u}_{0t} \otimes \mathbf{Z}_t) = \mathbf{0}_{dK_2 \times 1}$. Nevertheless, if the primary interest lies in estimating the time-varying long-run relationship $\mathbf{A}_t$, the representation in (17) provides guidance on how stationary variables may be incorporated to obtain a more tractable asymptotic distribution. Define the orthogonalized Brownian motion $\mathbf{B}_{0.z}(r) = \mathbf{B}_0(r) - \mathbf{\Sigma}_{0z}\mathbf{\Sigma}_{zz}^{-1}\mathbf{B}_Z(r)$. Then

$$\begin{aligned}\mathbb{E}\left[\mathbf{B}_{0.z}(r)\mathbf{B}_X^\top(r)\right] &= \mathbb{E}\left[\mathbf{B}_0(r)\mathbf{B}_X^\top(r)\right] - \mathbf{\Sigma}_{0z}\mathbf{\Sigma}_{zz}^{-1}\mathbb{E}\left[\mathbf{B}_Z(r)\mathbf{B}_X^\top(r)\right] \\ &= \mathbf{\Omega}_{0x} - \mathbf{\Sigma}_{0z}\mathbf{\Sigma}_{zz}^{-1}\mathbf{\Omega}_{zx} \\ &= \mathbf{\Sigma}_{0x} + \mathbf{\Lambda}_{0x} + \mathbf{\Lambda}_{x0}^\top - \mathbf{\Sigma}_{0z}\mathbf{\Sigma}_{zz}^{-1}\left(\mathbf{\Sigma}_{zx} + \mathbf{\Lambda}_{zx} + \mathbf{\Lambda}_{xz}^\top\right) \\ &= \mathbf{\Sigma}_{0x} - \mathbf{\Sigma}_{0z}\mathbf{\Sigma}_{zz}^{-1}\mathbf{\Sigma}_{zx} + \mathbf{\Lambda}_{0x} - \mathbf{\Sigma}_{0z}\mathbf{\Sigma}_{zz}^{-1}\mathbf{\Lambda}_{zx} + \mathbf{\Lambda}_{x0}^\top - \mathbf{\Sigma}_{0z}\mathbf{\Sigma}_{zz}^{-1}\mathbf{\Lambda}_{xz}^\top.\end{aligned}$$

A sufficient condition for the asymptotic distribution to be mixed normal is therefore $\mathbb{E}\left[\mathbf{B}_{0.z}(r)\mathbf{B}_X^\top(r)\right] = \mathbf{0}_{d\times K_1}$. To interpret this condition in terms of the underlying innovations, define the orthogonalized residual $\overline{\mathbf{u}}_{0,t} = \mathbf{u}_{0t} - \mathbf{\Sigma}_{0z}\mathbf{\Sigma}_{zz}^{-1}\mathbf{Z}_t$. Then, the preceding covariance components admit the representations

$$\mathbf{\Sigma}_{0x} - \mathbf{\Sigma}_{0z}\mathbf{\Sigma}_{zz}^{-1}\mathbf{\Sigma}_{zx} + \mathbf{\Lambda}_{0x} - \mathbf{\Sigma}_{0z}\mathbf{\Sigma}_{zz}^{-1}\mathbf{\Lambda}_{zx} = \sum_{j=0}^{\infty} \mathbb{E}\left(\overline{\mathbf{u}}_{0,t}\mathbf{u}_{x,t-j}^\top\right), \tag{22}$$

$$\mathbf{\Lambda}_{x0}^\top - \mathbf{\Sigma}_{0z}\mathbf{\Sigma}_{zz}^{-1}\mathbf{\Lambda}_{xz}^\top = \sum_{j=0}^{\infty} \mathbb{E}\left(\overline{\mathbf{u}}_{0,t}\mathbf{u}_{x,t+j}^\top\right). \tag{23}$$

Achieving a mixed normal limiting distribution requires both (22) and (23) to vanish. Consider a simple case where $\mathbf{\Lambda}_{0x} = \mathbf{\Lambda}_{zx} = \mathbf{0}$, In this case, by simply choosing $\mathbf{Z}_t = \Delta \mathbf{X}_t$ would make (22) vanish, but still not enough to kill (23). The structure of (22) and (23), nevertheless, motivates us to consider the following model specification by including both leads and lags of $\Delta \mathbf{X}_t = \mathbf{u}_{xt}$

$$\mathbf{y}_t = \mathbf{B}^{(k)} \mathbf{X}_t^{(k)} + \sum_{j=-K}^{K} \mathbf{\Gamma}_j \mathbf{X}_{t-j} + \widetilde{\mathbf{v}}_t^{(k)}, \tag{24}$$

where K is the number of leads (lags), $\widetilde{\mathbf{v}}_t^{(k)} = \mathbf{v}_t + \sum_{|j|>K} \mathbf{\Gamma}_j \mathbf{X}_{t-j} + (\mathbf{A}_t - \mathbf{A}_k(\tau_t)) \mathbf{X}_t$, and $(\mathbf{v}_t)_t$ is a stationary process satisfying $\mathbb{E}\left[\mathbf{u}_{xt} \mathbf{v}_{t+j}^\top\right] = \mathbf{0}_{K_1 \times d}$ for all j. The detailed justification of the specification (24) is provided in Section B.

We now present the following theorem for the series estimator $\widehat{\mathbf{A}}_n^{(k)}(\tau_t) = \widetilde{\mathbf{B}}_n^{(k)} \mathbf{T}_{(k)}(\tau_t)$, where $\widetilde{\mathbf{B}}_n^{(k)}$ is obtained from the LS estimation of (24).

Theorem 4. Suppose that Assumptions 1-3 are satisfied. If, in addition, $\frac{K^2}{n^{1-\gamma}} \to 0$ and $\sqrt{n} \sum_{|j|>K} \|\mathbf{\Gamma}_j\| \to 0$, where γ is defined as in Assumption 3. For any given $\tau_t \in [0,1]$, we have, as $n \to \infty$

$$n \left(\widetilde{\mathbb{V}}_{n(k)}(\tau_t)\right)^{-1/2} \operatorname{vec}\left(\widehat{\mathbf{A}}_n^{(k)}(\tau_t) - \mathbf{A}(\tau_t) + \mathbf{\Gamma}(\tau_t)\right) \xrightarrow{d} \mathcal{N}\left(\mathbf{0}_{dK_1 \times 1}, \mathbf{I}_{dK_1}\right), \tag{25}$$

where

$$\widetilde{\mathbb{V}}_{n(k)}(\tau_t) := \left\{\mathbf{T}_{(k)}^\top(\tau_t) \left(\mathbf{\Lambda}_{n,xx}^{(k)}\right)^{-1} \mathbf{T}_{(k)}(\tau_t)\right\} \otimes \widehat{\mathbf{\Omega}}_{n,0\cdot x},$$

and $\widehat{\mathbf{\Omega}}_{n,0\cdot x}$ is a consistent estimator of $\mathbf{\Omega}_{0\cdot x} = \mathbf{\Omega}_{00} - \mathbf{\Omega}_{0x} \mathbf{\Omega}_{xx}^{-1} \mathbf{\Omega}_{x0}$.

Theorem 4 shows that asymptotic distribution of the estimator is mixed Normal in the general model setup (1)-(2) , provided that enough number of leads and lags of $\Delta \mathbf{X}_t$ are augmented into the model. The requirement $\sqrt{n} \sum_{|j|>K} \|\mathbf{\Gamma}_j\| \to 0$ implies that K has to be large, so the approximation error becomes asymptotically negligible. In contrast, $\frac{K^2}{n^{1-\gamma}} \to 0$ restricts the rate at which K may diverge. In practice, some experiments show that a small value of K may be advisable, as detailed in Section 4. It may be noted, however, same problem also appears in the canonical cointegration regression approach proposed by Park and Hahn 1999, where the truncation lag parameter must also be selected in constructing HAC-type estimators for the long-run covariance matrix.

4 — Monte Carlo simulations

We use the Data Generating Process (D.G.P.):

$$y_t = A_t X_t + u_{0t},$$

where A_t has the following functional forms

- M1: $A_t = 1 + \frac{t}{n}$;
- M2: $A_t = e^{-\frac{t}{n}}$.

The regressor X_t is generated according to $X_t = X_{t-1} + u_{xt}$. We generate $\mathbf{u}_t = (u_{0t}, u_{xt})^\top$ from a bivariate MA(1) process

$$\mathbf{u}_t = \boldsymbol{\varepsilon}_t + \boldsymbol{\Theta}\boldsymbol{\varepsilon}_{t-1}, \quad \boldsymbol{\varepsilon}_t \overset{iid}{\sim} \mathcal{N}(0, \boldsymbol{\Sigma}_\varepsilon)$$

with

$$\boldsymbol{\Theta} = \begin{pmatrix} 0.3 & -0.4 \\ \theta_{21} & 0.6 \end{pmatrix}, \qquad \boldsymbol{\Sigma}_\varepsilon = \begin{pmatrix} 1 & \sigma_{21} \\ \sigma_{21} & 1 \end{pmatrix},$$

and θ_{21}, σ_{21} allowed to vary. We consider values of $\{0.8, 0.4, 0, -0.8\}$ for θ_{21} and $\{-0.85, -0.5, 0.5\}$ for σ_{21}, as in Phillips and Loretan 1991.

To compute the series estimator, we consider both the baseline specification

$$y_t = A_t X_t + u_{0t} \tag{26}$$

and a slightly more flexible specification in which number of lags (K_a) and leads (K_e) may be different:

$$y_t = A_t X_t + \sum_{j=-K_e}^{K_a} \gamma_j \Delta X_{t-j} + u_{0t}. \tag{27}$$

Following Phillips and Loretan 1991, we consider several combinations of the numbers of lags (K_a) and leads (K_e): (i) $K_a = K_e = 0$; (ii) $K_a = 2, K_e = 0$; (iii) $K_a = 4, K_e = 0$; (iv) $K_a = 2, K_e = 1$; and (v) $K_a = 4, K_e = 2$.[1]

1. Note that, in all these specifications based on (27), ΔX_t is always included in the model.

Regarding to the truncation parameter k, we use the generalized cross validation (GCV) method as in Gao, Tong, and Wolff 2002 to select an optima value k:

$$\hat{k} = \underset{k \in \mathcal{K}_n}{\arg\min} \left(1 - \frac{df(k, K_1, K_2)}{n}\right)^{-2} \frac{1}{n} \sum_{t=1}^{n} \left(\hat{u}_{0t}^{(k)}\right)^2,$$

where $df(k, K_1, K_2)$ is the total number of parameters to be estimated, $\mathcal{K}_n = \{1, 2, \cdots, 10, 11, 12\}$, and $\hat{u}_{0t}^{(m)}$ is the fitted residuals, either from (26) or (27).

The performance of the estimators is evaluated by the mean squared error (MSE): $\frac{1}{n}\sum_{t=1}^{n}\left(\hat{A}_t - A_t\right)^2$. The objective is to assess the accuracy of estimating the time-varying parameters A_t and, in particular, to examine whether augmenting the regression with leads and lags improves estimation accuracy.

Tables 1-2 report the average MSE of the series estimator of A_t under designs M1 and M2, respectively. For each combination of $(\theta_{21}, \sigma_{21})$ and sample size n, we compare the baseline specification (26) with the augmented specification (27) across a range of choices for the numbers of lags (K_a) and leads (K_e). Under (26), the reported MSEs are in levels, whereas all other entries are ratios relative to the baseline performance. Values below 1 indicate an improvement from augmentation, and boxed entries mark the best-performing specification.

Several patterns emerge. First, the gains from (27) are substantially larger when $\sigma_{21} < 0$. This is consistent with our DGP, under which $\Omega_{0x} = \theta_{21}(1.3 - 0.4\sigma_{21}) + (2.08\sigma_{21} - 0.64)$, which is larger in magnitude when σ_{21} is negative. Second, augmenting the model with leads of ΔX_t generally provides additional improvements: specifications including leads deliver the best performance in 31 out of 36 cases under M1 and 29 out of 36 cases under M2. Finally, the qualitative conclusions are highly stable across M1 and M2, indicating robustness to the functional form of the time-varying coefficient. Overall, in line with the simulation evidence, these results suggest that modest lead–lag augmentation is a robust way to improve estimation accuracy, especially in situations with stronger endogeneity.

5 — An empirical application

6 — Conclusion

Table 1—Small sample properties of the series estimator of A_t under M1: $A_t = 1 + \frac{t}{n}$

	$n = 200$				$n = 500$				$n = 1000$			
θ_{21}	0.8	0.4	0.0	-0.8	0.8	0.4	0.0	-0.8	0.8	0.4	0.0	-0.8
						$\sigma_{21} = -0.85$						
OLS	0.030	0.014	0.007	0.002	0.007	0.003	0.001	0.000	0.002	0.001	0.000	0.000
(0,0)	0.418	0.242	0.158	0.172	0.398	0.235	0.168	0.204	0.405	0.246	0.170	0.290
(2,0)	0.246	0.194	0.155	0.147	0.217	0.186	0.158	0.168	0.209	0.181	0.171	0.252
(4,0)	0.240	0.202	0.150	0.131	0.218	$\boxed{0.175}$	0.159	0.165	0.210	0.181	0.170	0.237
(2,1)	0.246	0.204	0.155	0.115	$\boxed{0.204}$	0.188	0.161	0.137	$\boxed{0.189}$	0.176	0.167	0.215
(4,2)	0.242	0.195	0.155	0.106	0.209	0.184	$\boxed{0.151}$	0.116	0.194	0.173	0.171	0.185
(1,1)	$\boxed{0.231}$	0.199	0.149	0.122	0.215	0.186	0.167	0.135	0.206	0.188	0.172	0.215
(4,4)	0.248	$\boxed{0.192}$	$\boxed{0.143}$	$\boxed{0.095}$	0.205	0.184	0.159	0.109	0.200	0.174	0.163	0.175
(8,8)	0.269	0.220	0.152	0.097	0.211	0.185	0.152	$\boxed{0.098}$	0.198	$\boxed{0.163}$	$\boxed{0.157}$	$\boxed{0.159}$
						$\sigma_{21} = -0.5$						
OLS	0.010	0.009	0.005	0.002	0.002	0.002	0.001	0.000	0.001	0.000	0.000	0.000
(0,0)	0.729	0.598	0.473	0.370	0.741	0.588	0.480	0.451	0.715	0.571	0.492	0.495
(2,0)	0.585	0.515	$\boxed{0.454}$	0.387	0.567	0.537	0.450	0.443	0.535	0.503	0.482	0.520
(4,0)	0.620	$\boxed{0.510}$	0.458	0.374	0.557	0.520	0.436	0.457	0.536	0.491	0.455	0.498
(2,1)	0.589	0.511	0.482	0.342	$\boxed{0.525}$	0.514	$\boxed{0.432}$	0.396	0.491	0.497	$\boxed{0.448}$	0.450
(4,2)	0.565	0.550	0.454	$\boxed{0.312}$	0.549	$\boxed{0.503}$	0.456	0.369	0.494	$\boxed{0.453}$	0.482	0.414
(1,1)	$\boxed{0.554}$	0.517	0.455	0.334	0.554	0.523	0.442	0.389	0.496	0.524	0.483	0.432
(4,4)	0.643	0.535	0.463	0.331	0.539	0.504	0.442	0.344	$\boxed{0.466}$	0.463	0.449	$\boxed{0.399}$
(8,8)	0.690	0.599	0.525	0.323	0.528	0.520	0.432	$\boxed{0.327}$	0.495	0.467	0.454	0.374
						$\sigma_{21} = 0.5$						
OLS	0.001	0.002	0.002	0.003	0.000	0.000	0.000	0.001	0.000	0.000	0.000	0.000
(0,0)	1.297	1.204	1.200	1.075	1.187	1.285	1.174	0.897	1.167	1.125	1.237	0.925
(2,0)	1.035	1.010	1.114	1.003	0.906	1.021	1.053	0.937	0.909	0.995	1.049	1.058
(4,0)	1.046	1.011	1.088	1.064	0.883	1.001	1.008	0.964	0.893	0.920	1.003	1.029
(2,1)	$\boxed{0.965}$	1.036	1.208	$\boxed{0.886}$	0.746	$\boxed{0.914}$	0.999	0.801	0.728	0.851	0.986	0.856
(4,2)	0.972	1.030	1.147	0.917	$\boxed{0.735}$	0.944	0.997	0.791	0.690	0.825	1.001	0.821
(1,1)	0.965	1.022	1.113	0.893	0.777	0.975	1.089	0.797	0.786	0.927	1.081	0.806
(4,4)	0.986	1.040	1.176	0.892	0.740	0.943	0.992	0.802	0.695	$\boxed{0.792}$	$\boxed{0.970}$	0.847
(8,8)	1.118	1.165	1.204	0.941	0.753	0.942	$\boxed{0.978}$	$\boxed{0.785}$	$\boxed{0.625}$	0.819	0.985	$\boxed{0.786}$

Notes. OLS is based on specification (26). All other rows are based on specification (27), with different combinations of the numbers of leads K_e and lags K_a. Entries in the row "OLS" report MSE in levels, while the remaining entries are ratios relative to OLS performance. For each combination of $(\theta_{21}, \sigma_{21})$ and sample size n, boxed entries indicate the best performance.

Table 2 — Small sample properties of the series estimator of A_t under M2: $A_t = e^{-\frac{t}{n}}$

	$n = 200$				$n = 500$				$n = 1000$			
θ_{21}	0.8	0.4	0.0	-0.8	0.8	0.4	0.0	-0.8	0.8	0.4	0.0	-0.8
					$\sigma_{21} = -0.85$							
OLS	0.028	0.013	0.006	0.002	0.006	0.003	0.001	0.000	0.002	0.001	0.000	0.000
(0,0)	0.424	0.230	0.152	0.129	0.416	0.250	0.163	0.173	0.412	0.260	0.171	0.196
(2,0)	0.258	0.199	0.145	0.110	0.228	0.192	0.152	0.143	0.204	0.182	0.156	0.173
(4,0)	0.269	$\boxed{0.190}$	0.145	0.100	0.222	$\boxed{0.178}$	0.158	0.137	0.204	0.178	0.160	0.163
(2,1)	0.262	0.201	$\boxed{0.138}$	0.093	0.220	0.180	0.168	0.123	0.190	$\boxed{0.167}$	0.159	0.148
(4,2)	0.260	0.217	0.146	0.084	$\boxed{0.213}$	0.182	0.150	0.109	0.195	0.177	0.153	0.124
(1,1)	$\boxed{0.253}$	0.203	0.150	0.102	0.217	0.184	0.156	0.127	$\boxed{0.189}$	0.176	0.150	0.153
(4,4)	0.266	0.217	0.152	$\boxed{0.081}$	0.214	0.186	0.155	0.107	0.194	0.171	0.153	0.118
(8,8)	0.313	0.243	0.164	0.089	0.223	0.190	$\boxed{0.148}$	$\boxed{0.090}$	0.197	0.168	$\boxed{0.149}$	$\boxed{0.107}$
					$\sigma_{21} = -0.5$							
OLS	0.010	0.008	0.005	0.002	0.002	0.002	0.001	0.000	0.001	0.000	0.000	0.000
(0,0)	0.764	0.602	0.444	0.311	0.735	0.528	0.407	0.372	0.761	0.589	0.480	0.409
(2,0)	0.640	0.561	$\boxed{0.405}$	0.318	0.556	0.455	0.395	0.372	0.548	0.506	0.444	0.398
(4,0)	0.692	$\boxed{0.531}$	0.418	0.292	0.582	0.446	0.406	0.348	0.556	0.471	0.415	0.383
(2,1)	0.614	0.566	0.439	0.291	0.524	0.461	0.406	0.340	0.508	0.465	0.437	0.363
(4,2)	$\boxed{0.608}$	0.594	0.428	$\boxed{0.282}$	$\boxed{0.506}$	$\boxed{0.446}$	0.405	0.307	$\boxed{0.487}$	$\boxed{0.461}$	0.432	0.327
(1,1)	0.619	0.562	0.407	0.293	0.534	0.470	0.421	0.321	0.527	0.486	0.443	0.341
(4,4)	0.627	0.599	0.451	0.294	0.530	0.462	0.424	0.307	0.513	0.463	$\boxed{0.414}$	0.322
(8,8)	0.715	0.645	0.483	0.313	0.565	0.479	$\boxed{0.393}$	$\boxed{0.302}$	0.517	0.487	0.430	$\boxed{0.305}$
					$\sigma_{21} = 0.5$							
OLS	0.001	0.001	0.002	0.003	0.000	0.000	0.000	0.001	0.000	0.000	0.000	0.000
(0,0)	1.204	1.146	1.281	0.975	1.262	1.207	1.158	0.991	1.259	1.218	1.170	0.992
(2,0)	0.956	0.995	1.233	0.991	0.972	0.997	1.012	1.027	0.967	0.953	1.028	1.022
(4,0)	$\boxed{0.953}$	$\boxed{0.986}$	1.230	0.991	0.959	0.937	1.068	0.986	0.874	0.928	1.014	0.992
(2,1)	1.008	1.098	1.274	$\boxed{0.858}$	0.840	0.938	1.047	0.860	0.751	0.854	0.994	0.875
(4,2)	1.039	1.105	1.258	0.883	0.831	0.965	1.053	0.888	0.718	0.883	0.962	0.858
(1,1)	0.973	1.035	1.255	0.864	0.881	0.942	1.104	0.871	0.768	0.918	1.044	0.886
(4,4)	1.083	1.119	1.297	0.906	$\boxed{0.783}$	$\boxed{0.933}$	1.069	$\boxed{0.834}$	0.705	0.842	0.999	$\boxed{0.831}$
(8,8)	1.130	1.197	1.407	1.033	0.811	0.944	1.041	0.844	$\boxed{0.661}$	$\boxed{0.837}$	$\boxed{0.960}$	0.840

Notes. See Notes in Table 1.

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A — Proofs of the main theorems

A.1. Proof of Theorem 1

Recall that the estimator for $\mathbf{B}^{(k)}$ is given by

$$\widehat{\mathbf{B}}_n^{(k)} = \left(\sum_{t=1}^n \mathbf{y}_t \mathbf{X}_t^{(k)}\right)\left(\sum_{t=1}^n \mathbf{X}_t^{(k)}\mathbf{X}_t^{(k)\top}\right)^{-1}.$$

Simple algebra gives

$$\begin{aligned} n\cdot\left(\widehat{\mathbf{B}}_n^{(k)} - \mathbf{B}^{(k)}\right) &= \left(\frac{1}{n}\sum_{t=1}^n \mathbf{u}_{0t}^{(k)}\mathbf{X}_t^{(k)\top}\right)\left(\frac{1}{n^2}\sum_{t=1}^n \mathbf{X}_t^{(k)}\mathbf{X}_t^{(k)\top}\right)^{-1} \\ &= \left(\frac{1}{n}\sum_{t=1}^n \mathbf{u}_{0t}\mathbf{X}_t^{(k)\top}\right)\left(\frac{1}{n^2}\sum_{t=1}^n \mathbf{X}_t^{(k)}\mathbf{X}_t^{(k)\top}\right)^{-1} \\ &\qquad + \left(\frac{1}{n}\sum_{t=1}^n \left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\mathbf{X}_t\mathbf{X}_t^{(k)\top}\right)\left(\frac{1}{n^2}\sum_{t=1}^n \mathbf{X}_t^{(k)}\mathbf{X}_t^{(k)\top}\right)^{-1} \\ &:= \mathbf{\Gamma}_{n,ux}^{(k)}\left(\mathbf{\Lambda}_{n,xx}^{(k)}\right)^{-1} + \mathbf{\Delta}_{n,xx}^{(k)}\left(\mathbf{\Lambda}_{n,xx}^{(k)}\right)^{-1}. \end{aligned}$$

By Lemma C.1, we have

$$\mathbf{\Lambda}_{n,xx}^{(k)} = \mathbf{\Lambda}_{xx}^{(k)} + o_p(1),$$

where $\mathbf{\Lambda}_{xx}^{(k)} = \int_0^1 \left(\boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r)\right)\otimes\left(\mathbf{B}_X(r)\mathbf{B}_X^\top(r)\right)dr$ and $\left\|\left(\mathbf{\Lambda}_{n,xx}^{(k)}\right)^{-1}\right\|_{sp} = O_p(1)$. In addition, Lemma C.3 leads to

$$\frac{1}{n}\mathbf{\Delta}_{n,xx}^{(k)} = \mathbf{\Delta}_{xx}^{(k)} + o_p(1),$$

where $\left\|\mathbf{\Delta}_{xx}^{(k)}\right\|_{sp} = O_p\left(\frac{1}{k^q}\right)$. This implies that $\left\|\mathbf{\Delta}_{n,xx}^{(k)}\left(\mathbf{\Lambda}_{n,xx}^{(k)}\right)^{-1}\right\|_{sp} \le \left\|\mathbf{\Delta}_{xx}^{(k)}\right\|_{sp}\left\|\left(\mathbf{\Lambda}_{n,xx}^{(k)}\right)^{-1}\right\|_{sp} = O_p\left(\frac{n}{k^q}\right) = o_p(1)$.

Define $\mathbf{\Gamma}(\tau_t) := \mathbf{A}(\tau_t) - \mathbf{A}_k(\tau_t)$ and $\mathbf{T}_{(k)}(\tau_t) = \boldsymbol{\psi}_{(k)}(\tau_t)\otimes\mathbf{I}_{K_1}$ for any given $\tau_t = t/n$. Then, we have

$$
\begin{aligned}
n \cdot \left(\widehat{\mathbf{A}}_n^{(k)}(\tau_t) - \mathbf{A}(\tau_t) + \boldsymbol{\Gamma}(\tau_t) \right) &= n \cdot \left[\left(\widehat{\mathbf{B}}_n^{(k)} - \mathbf{B}^{(k)} \right) \left(\boldsymbol{\psi}_{(k)}(\tau_t) \otimes \mathbf{I}_{K_1} \right) \right] \\
&= \boldsymbol{\Gamma}_{ux}^{(k)} \boldsymbol{\Lambda}_{xx}^{(k)-1} \mathbf{T}_{(k)}(\tau_t) + o_p(\sqrt{k}),
\end{aligned}
$$

where $\boldsymbol{\Gamma}_{ux}^{(k)} = \int_0^1 d\mathbf{B}_0(r) \left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r) \right)^\top + \boldsymbol{\ell}_{1(k)} \otimes (\boldsymbol{\Lambda}_{0x} + \boldsymbol{\Sigma}_{0x})$, which follows from Lemma C.1 and Lemma C.2.

A.2. Proof of Corollary 1

Let

$$
\widehat{\mathbb{V}}_{n(k)}(\tau_t) = \left\{ \mathbf{T}_{(k)}^\top(\tau_t) \left(\boldsymbol{\Lambda}_{n,xx}^{(k)} \right)^{-1} \mathbf{T}_{(k)}(\tau_t) \right\} \otimes \widehat{\boldsymbol{\Omega}}_{n,00},
$$

where $\widehat{\boldsymbol{\Omega}}_{n,00}$ is a consistent estimator of $\boldsymbol{\Omega}_{00}$. Then, we have

$$
\begin{aligned}
&n \left(\widehat{\mathbb{V}}_{n(k)}(\tau_t) \right)^{-1/2} \operatorname{vec} \left(\widehat{\mathbf{A}}_n^{(k)}(\tau_t) - \mathbf{A}(\tau_t) + \boldsymbol{\Gamma}(\tau_t) \right) \\
&= \left(\left\{ \mathbf{T}_{(k)}^\top(u) \left(\boldsymbol{\Lambda}_{n,xx}^{(k)} \right)^{-1} \mathbf{T}_{(k)}(u) \right\} \otimes \widehat{\boldsymbol{\Omega}}_{n,00} \right)^{-1/2} \left(\mathbf{T}_{(k)}^\top(\tau_t) \otimes \mathbf{I}_d \right) \operatorname{vec} \left\{ \boldsymbol{\Gamma}_{n,ux}^{(k)} \left(\boldsymbol{\Lambda}_{n,xx}^{(k)} \right)^{-1} \right\} + o_p(1).
\end{aligned}
$$

Using the joint convergence result of

$$
\begin{pmatrix} \frac{1}{\sqrt{n}} \sum_{t=1}^{\lfloor nr \rfloor} \mathbf{u}_{0t} \\ \frac{1}{\sqrt{n}} \sum_{t=1}^{\lfloor nr \rfloor} \mathbf{u}_{xt} \end{pmatrix} \Rightarrow \begin{pmatrix} \mathbf{B}_0(r) \\ \mathbf{B}_X(r) \end{pmatrix},
$$

on $D\big([0,1],\, \mathbb{R}^{d+K_1}\big)$, together with Lemma C.1 and Lemma C.2, we have the joint stochastic expansion

$$
\begin{pmatrix} \boldsymbol{\Lambda}_{n,xx}^{(k)} \\ \boldsymbol{\Gamma}_{n,ux}^{(k)} \end{pmatrix} = \begin{pmatrix} \boldsymbol{\Lambda}_{xx}^{(k)} \\ \boldsymbol{\Gamma}_{ux}^{(k)} \end{pmatrix} + o_p(1),
$$

where $\boldsymbol{\Lambda}_{xx}^{(k)}$ and $\boldsymbol{\Gamma}_{ux}^{(k)}$ are defined as in Theorem 1. Since $\mathbb{E}\left(\mathbf{u}_{0t} \otimes \mathbf{u}_{xs} \right) = \mathbf{0}_{dK_1 \times 1}$ for any t, s, we have $\boldsymbol{\Omega}_{0x} = \mathbf{0}_{dK_1}$, so that Brownian motions $\mathbf{B}_0(r)$ and $\mathbf{B}_X(r)$ are uncorrelated and hence independently distributed. Then, as in Phillips and Hansen 1990, together with the continuous mapping theorem, we can show that

$$\left\{\mathbf{T}_{(k)}^{\top}(\tau_t)\otimes\mathbf{I}_d\right\}\left\{\left(\mathbf{\Lambda}_{xx}^{(k)}\right)^{-1}\otimes\mathbf{I}_d\right\}\int_0^1\left(\left(\boldsymbol{\psi}_{(k)}(r)\otimes\mathbf{B}_X(r)\right)\otimes\mathbf{I}_d\right)d\mathbf{B}_r(\mathbf{\Omega}_{00})$$

has a mixed Normal distribution with stochastic covariance matrix $\mathbb{V}_{(k)}(\tau_t):=\left\{\mathbf{T}_{(k)}^{\top}(\tau_t)\left(\mathbf{\Lambda}_{xx}^{(k)}\right)^{-1}\mathbf{T}_{(k)}(\tau_t)\right\}\otimes\mathbf{\Omega}_{00}$. Clearly, we have

$$\left\|\mathbb{V}_{(k)}(\tau)\right\|_{sp}\leq\left\|\mathbf{\Omega}_{00}\right\|_{sp}\left\|\left(\mathbf{\Lambda}_{xx}^{(k)}\right)^{-1}\right\|_{sp}\left\|\boldsymbol{\psi}_{(k)}(r)\right\|^2=O_p(k).$$

As $\widehat{\mathbf{\Omega}}_{n,00}\xrightarrow{p}\mathbf{\Omega}_{00}$ implies $\widehat{\mathbb{V}}_{n(k)}(\tau_t)=\mathbb{V}_{(k)}(\tau_t)+o_p(1)$, by Slutsky's theorem and continuous mapping theorem, we obtain

$$n\left(\widehat{\mathbb{V}}_{n(k)}(\tau_t)\right)^{-1/2}\operatorname{vec}\left(\widehat{\mathbf{A}}_n^{(k)}(\tau_t)-\mathbf{A}(\tau_t)+\mathbf{\Gamma}(\tau_t)\right)\xrightarrow{d}\mathcal{N}\left(\mathbf{0}_{dK_1\times 1},\mathbf{I}_{dK_1}\right).$$

A.3. Proof of Theorem 2

Define

$$\begin{aligned}\widehat{\mathbf{B}}_X&=\left(\sum_{t=1}^n\mathbf{y}_t\mathbf{X}_t^{(k)\top}\right)\left(\sum_{t=1}^n\mathbf{X}_t^{(k)}\mathbf{X}_t^{(k)\top}\right)^{-1},\\ \widehat{\mathbf{\Gamma}}_Z&=\left(\sum_{t=1}^n\mathbf{Z}_t\mathbf{X}_t^{(k)\top}\right)\left(\sum_{t=1}^n\mathbf{X}_t^{(k)}\mathbf{X}_t^{(k)\top}\right)^{-1}.\end{aligned}$$

Then,

$$\widetilde{\mathbf{y}}_t=\mathbf{y}_t=\widehat{\mathbf{B}}_X\mathbf{X}_t^{(k)},\quad\widetilde{\mathbf{Z}}_t=\mathbf{Z}_t-\widehat{\mathbf{\Gamma}}_Z\mathbf{X}_t^{(k)},$$

are just residuals from the regression $\mathbf{y}_t$ on $\mathbf{X}_t^{(k)}$ and $\mathbf{Z}_t$ on $\mathbf{X}_t^{(k)}$, respectively. In view of the Frisch–Waugh–Lovell theorem, $\mathbf{\Gamma}$ can be estimated from the regression

$$\widetilde{\mathbf{y}}_t=\mathbf{\Gamma}\widetilde{\mathbf{Z}}_t+\widetilde{\mathbf{u}}_{0t}^{(k)},$$

where $\widetilde{\mathbf{u}}_{0t}^{(k)} = \mathbf{u}_{0t} + (\mathbf{A}_t - \mathbf{A}_k(t/n))\,\mathbf{X}_t + \left(\mathbf{B}^{(k)} - \widehat{\mathbf{B}}_X + \mathbf{\Gamma}\widehat{\mathbf{\Gamma}}_Z\right)\mathbf{X}_t^{(k)}$. We then have

$$\begin{aligned}
\widehat{\mathbf{\Gamma}}_n - \mathbf{\Gamma} &= \left(\sum_{t=1}^{n} \widetilde{\mathbf{u}}_{0t}^{(k)} \widetilde{\mathbf{Z}}_t^\top\right) \left(\sum_{t=1}^{n} \widetilde{\mathbf{Z}}_t \widetilde{\mathbf{Z}}_t^\top\right)^{-1} \\
&= \left(\sum_{t=1}^{n} \mathbf{u}_{0t} \widetilde{\mathbf{Z}}_t^\top\right) \left(\sum_{t=1}^{n} \widetilde{\mathbf{Z}}_t \widetilde{\mathbf{Z}}_t^\top\right)^{-1} + \left(\sum_{t=1}^{n} (\mathbf{A}_t - \mathbf{A}_k(t/n))\,\mathbf{X}_t \widetilde{\mathbf{Z}}_t^\top\right) \left(\sum_{t=1}^{n} \widetilde{\mathbf{Z}}_t \widetilde{\mathbf{Z}}_t^\top\right)^{-1},
\end{aligned}$$

since, by construction, $\sum_{t=1}^{n} \mathbf{X}_t^{(k)} \widetilde{\mathbf{Z}}_t^\top = \mathbf{0}_{K_1 \times K_2}$.

First, we note that

$$\frac{1}{n}\sum_{t=1}^{n} \widetilde{\mathbf{Z}}_t \widetilde{\mathbf{Z}}_t^\top = \frac{1}{n}\sum_{t=1}^{n} \mathbf{Z}_t \mathbf{Z}_t^\top - \left(\frac{1}{n}\sum_{t=1}^{n} \mathbf{Z}_t \mathbf{X}_t^{(k)\top}\right) \left(\frac{1}{n}\sum_{t=1}^{n} \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top}\right)^{-1} \left(\frac{1}{n}\sum_{t=1}^{n} \mathbf{X}_t^{(k)} \mathbf{Z}_t^\top\right).$$

By Lemma C.1, we have $\left\|\left(\frac{1}{n}\sum_{t=1}^{n} \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top}\right)^{-1}\right\|_{sp} = O_p(1/n)$. In addition, Lemma C.4(i) leads to $\left\|\frac{1}{n}\sum_{t=1}^{n} \mathbf{X}_t^{(k)} \mathbf{Z}_t^\top\right\|_{sp} = O_p(1)$. This implies that

$$\begin{aligned}
\left\|\frac{1}{n}\sum_{t=1}^{n} \widetilde{\mathbf{Z}}_t \widetilde{\mathbf{Z}}_t^\top - \frac{1}{n}\sum_{t=1}^{n} \mathbf{Z}_t \mathbf{Z}_t^\top\right\|_{sp} &\le \left\|\left(\frac{1}{n}\sum_{t=1}^{n} \mathbf{Z}_t \mathbf{X}_t^{(k)\top}\right) \left(\frac{1}{n}\sum_{t=1}^{n} \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top}\right)^{-1} \left(\frac{1}{n}\sum_{t=1}^{n} \mathbf{X}_t^{(k)} \mathbf{Z}_t^\top\right)\right\|_{sp} \\
&\le \left\|\frac{1}{n}\sum_{t=1}^{n} \mathbf{Z}_t \mathbf{X}_t^{(k)\top}\right\|_{sp} \left\|\left(\frac{1}{n}\sum_{t=1}^{n} \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top}\right)^{-1}\right\|_{sp} \left\|\frac{1}{n}\sum_{t=1}^{n} \mathbf{X}_t^{(k)} \mathbf{Z}_t^\top\right\|_{sp} \\
&= O_p(1/n),
\end{aligned}$$

so that

$$\frac{1}{n}\sum_{t=1}^{n} \widetilde{\mathbf{Z}}_t \widetilde{\mathbf{Z}}_t^\top = \frac{1}{n}\sum_{t=1}^{n} \mathbf{Z}_t \mathbf{Z}_t^\top + O_p(1/n). \tag{A.1}$$

Similar arguments yield

$$\begin{aligned}
\frac{1}{\sqrt{n}}\sum_{t=1}^{n}\mathbf{u}_{0t}\widetilde{\mathbf{Z}}_t^{\top} &= \frac{1}{\sqrt{n}}\sum_{t=1}^{n}\mathbf{u}_{0t}\mathbf{Z}_t^{\top} - \frac{1}{\sqrt{n}}\sum_{t=1}^{n}\mathbf{u}_{0t}\mathbf{X}_t^{\top(k)}\widehat{\boldsymbol{\Gamma}}_Z^{\top} \\
&= \frac{1}{\sqrt{n}}\sum_{t=1}^{n}\mathbf{u}_{0t}\mathbf{Z}_t^{\top} - \left(\frac{1}{\sqrt{n}}\sum_{t=1}^{n}\mathbf{u}_{0t}\mathbf{X}_t^{\top(k)}\right)\left(\frac{1}{n}\sum_{t=1}^{n}\mathbf{X}_t^{(k)}\mathbf{X}_t^{(k)\top}\right)^{-1}\left(\frac{1}{n}\sum_{t=1}^{n}\mathbf{X}_t^{(k)}\mathbf{Z}_t^{\top}\right) \\
&= \frac{1}{\sqrt{n}}\sum_{t=1}^{n}\mathbf{u}_{0t}\mathbf{Z}_t^{\top} + O_p\left(1/\sqrt{n}\right).
\end{aligned} \tag{A.2}$$

We also have

$$\begin{aligned}
&\frac{1}{\sqrt{n}}\sum_{t=1}^{n}\left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\mathbf{X}_t\widetilde{\mathbf{Z}}_t^{\top} \\
&= \frac{1}{\sqrt{n}}\sum_{t=1}^{n}\left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\mathbf{X}_t\mathbf{Z}_t^{\top} - \frac{1}{\sqrt{n}}\sum_{t=1}^{n}\left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\mathbf{X}_t\mathbf{X}_t^{(k)\top}\widehat{\boldsymbol{\Gamma}}_Z^{\top} \\
&= \frac{1}{\sqrt{n}}\sum_{t=1}^{n}\left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\mathbf{X}_t\mathbf{Z}_t^{\top} - \left(\frac{1}{\sqrt{n}}\sum_{t=1}^{n}\left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\mathbf{X}_t\mathbf{X}_t^{(k)\top}\right)\left(\sum_{t=1}^{n}\mathbf{X}_t^{(k)}\mathbf{X}_t^{(k)\top}\right)^{-1}\left(\sum_{t=1}^{n}\mathbf{X}_t^{(k)}\mathbf{Z}_t^{\top}\right) \\
&= O_p\left(\frac{n}{k^q}\right) + O_p\left(\frac{n^{3/2}}{k^q}\right)O_p\left(\frac{1}{n}\right) = O_p\left(\frac{n}{k^q}\right),
\end{aligned} \tag{A.3}$$

where the third equality follows from Lemma C.4(ii). By combining (A.1), (A.2), and (A.3), we obtain

$$\sqrt{n}\cdot\left(\hat{\boldsymbol{\Gamma}}_n - \boldsymbol{\Gamma}\right) = \left(\frac{1}{\sqrt{n}}\sum_{t=1}^{n}\mathbf{u}_{0t}\mathbf{Z}_t^{\top}\right)\left(\frac{1}{n}\sum_{t=1}^{n}\mathbf{Z}_t\mathbf{Z}_t^{\top}\right)^{-1} + o_p(1).$$

Using standard WLLN and C.L.T. for stationary linear processes (see, for example, Lemma 8.4 in Phillips 1995), we may show that

$$\frac{1}{n}\sum_{t=1}^{n}\mathbf{Z}_t\mathbf{Z}_t^{\top} \xrightarrow{p} \boldsymbol{\Sigma}_{zz},$$

and

$$\frac{1}{\sqrt{n}}\sum_{t=1}^{n}\mathrm{vec}\left(\mathbf{u}_{0t}\mathbf{Z}_t^{\top}\right) \xrightarrow{d} \mathcal{N}\left(\mathbf{0}_{dK_2\times 1}, \boldsymbol{\Omega}^{(0z)}\right),$$

where $\boldsymbol{\Omega}^{(0z)} = \sum_{j=-\infty}^{\infty}\mathbb{E}\left(\mathbf{u}_{0t}\mathbf{u}_{0t+j}^{\top}\otimes\mathbf{Z}_t\mathbf{Z}_{t+j}^{\top}\right)$. Theorem 2 now follows immediately from Slutsky's theorem.

A.4. Proof of Theorem 3

Similarly as in the proof of Theorem 1, define

$$\widehat{\mathbf{B}}_{YZ} = \left(\sum_{t=1}^{n} \mathbf{y}_t \mathbf{Z}_t^\top\right)\left(\sum_{t=1}^{n} \mathbf{Z}_t \mathbf{Z}_t^\top\right)^{-1},$$
$$\widehat{\mathbf{\Gamma}}_{XZ} = \left(\sum_{t=1}^{n} \mathbf{X}_t^{(k)} \mathbf{Z}_t^\top\right)\left(\sum_{t=1}^{n} \mathbf{Z}_t \mathbf{Z}_t^\top\right)^{-1}.$$

Then,

$$\check{\mathbf{y}}_t = \mathbf{y}_t - \widehat{\mathbf{B}}_{YZ}\mathbf{Z}_t, \quad \check{\mathbf{X}}_t^{(k)} = \mathbf{X}_t^{(k)} - \widehat{\mathbf{\Gamma}}_{XZ}\mathbf{Z}_t,$$

are just residuals from the regression $\mathbf{y}_t$ on $\mathbf{Z}_t$ and $\mathbf{X}_t^{(k)}$ on $\mathbf{Z}_t$, respectively. In view of the Frisch–Waugh–Lovell theorem, $\mathbf{B}^{(k)}$ can be estimated from the regression

$$\check{\mathbf{y}}_t = \mathbf{B}^{(k)}\check{\mathbf{X}}_t^{(k)} + \check{\mathbf{u}}_{0t}^{(k)},$$

where $\check{\mathbf{u}}_{0t}^{(k)} = \mathbf{u}_{0t} + \left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\mathbf{X}_t + \left(\mathbf{B}^{(k)}\widehat{\mathbf{\Gamma}}_{XZ} - \widehat{\mathbf{B}}_{YZ} + \mathbf{\Gamma}\right)\mathbf{Z}_t$. Then, we have

$$\begin{aligned}\widehat{\mathbf{B}}_n^{(k)} - \mathbf{B}^{(k)} &= \left(\sum_{t=1}^{n} \check{\mathbf{u}}_{0t}^{(k)} \check{\mathbf{X}}_t^{(k)\top}\right)\left(\sum_{t=1}^{n} \check{\mathbf{X}}_t^{(k)} \check{\mathbf{X}}_t^{(k)\top}\right)^{-1} \\ &= \left(\sum_{t=1}^{n} \mathbf{u}_{0t} \check{\mathbf{X}}_t^{(k)\top}\right)\left(\sum_{t=1}^{n} \check{\mathbf{X}}_t^{(k)} \check{\mathbf{X}}_t^{(k)\top}\right)^{-1} + \left(\sum_{t=1}^{n} \left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\mathbf{X}_t \check{\mathbf{X}}_t^{(k)\top}\right)\left(\sum_{t=1}^{n} \check{\mathbf{X}}_t^{(k)} \check{\mathbf{X}}_t^{(k)\top}\right)^{-1},\end{aligned} \tag{A.4}$$

since, by construction, $\sum_{t=1}^{n} \check{\mathbf{X}}_t^{(k)} \mathbf{Z}_t^\top = \mathbf{0}_{K_1 \times K_2}$.

Proceeding analogously as in the proof of Theorem 2, we have

$$\begin{aligned}\frac{1}{n^2}\sum_{t=1}^{n} \check{\mathbf{X}}_t^{(k)} \check{\mathbf{X}}_t^{(k)\top} &= \frac{1}{n^2}\sum_{t=1}^{n} \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} + O_p\left(\frac{1}{n}\right) \\ &= \int_0^1 \left\{\boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r) \otimes \left(\mathbf{B}_X(r)\mathbf{B}_X^\top(r)\right)\right\} dr + o_p(1).\end{aligned} \tag{A.5}$$

Second, observe that

$$\frac{1}{n}\sum_{t=1}^{n}\mathbf{u}_{0t}\check{\mathbf{X}}_t^{(k)\top}=\frac{1}{n}\sum_{t=1}^{n}\mathbf{u}_{0t}\mathbf{X}_t^{(k)\top}-\left(\frac{1}{n}\sum_{t=1}^{n}\mathbf{u}_{0t}\mathbf{Z}_t^{\top}\right)\left(\frac{1}{n}\sum_{t=1}^{n}\mathbf{Z}_t\mathbf{Z}_t^{\top}\right)^{-1}\left(\frac{1}{n}\sum_{t=1}^{n}\mathbf{Z}_t\mathbf{X}_t^{(k)\top}\right).$$

Standard WLLN for stationary linear processes gives

$$\frac{1}{n}\sum_{t=1}^{n}\mathbf{u}_{0t}\mathbf{Z}_t^{\top}\xrightarrow{p}\mathbf{\Sigma}_{0z},\quad\left(\frac{1}{n}\sum_{t=1}^{n}\mathbf{Z}_t\mathbf{Z}_t^{\top}\right)^{-1}\xrightarrow{p}\mathbf{\Sigma}_{zz}^{-1}.$$

Recall that, in Lemma C.2 and Lemma C.4(i), we have shown

$$\begin{aligned}\frac{1}{n}\sum_{t=1}^{n}\mathbf{u}_{0t}\mathbf{X}_t^{(k)\top}&=\int_0^1 d\mathbf{B}_0(r)\left(\boldsymbol{\psi}_{(k)}(r)\otimes\mathbf{B}_X(r)\right)^{\top}+\boldsymbol{\ell}_{1(k)}^{\top}\otimes\left(\mathbf{\Lambda}_{0x}+\mathbf{\Sigma}_{0x}\right)+o_p(1),\\ \frac{1}{n}\sum_{t=1}^{n}\mathbf{Z}_t\mathbf{X}_t^{(k)\top}&=\int_0^1 d\mathbf{B}_Z(r)\left(\boldsymbol{\psi}_{(k)}(r)\otimes\mathbf{B}_X(r)\right)^{\top}+\boldsymbol{\ell}_{1(k)}^{\top}\otimes\left(\mathbf{\Lambda}_{zx}+\mathbf{\Sigma}_{zx}\right)+o_p(1).\end{aligned}$$

By combining results above, we obtain

$$\begin{aligned}\frac{1}{n}\sum_{t=1}^{n}\mathbf{u}_{0t}\check{\mathbf{X}}_t^{(k)\top}=&\int_0^1\left(d\mathbf{B}_0(r)-\mathbf{\Sigma}_{0z}\mathbf{\Sigma}_{zz}^{-1}d\mathbf{B}_X(r)\right)\left(\boldsymbol{\psi}_{(k)}(r)\otimes\mathbf{B}_X(r)\right)^{\top}\\&+\boldsymbol{\ell}_{1(k)}^{\top}\otimes\left\{\mathbf{\Lambda}_{0x}+\mathbf{\Sigma}_{0x}-\mathbf{\Sigma}_{0z}\mathbf{\Sigma}_{zz}^{-1}\left(\mathbf{\Lambda}_{zx}+\mathbf{\Sigma}_{zx}\right)\right\}+o_p(1).\end{aligned}\tag{A.6}$$

Finally,

$$\begin{aligned}&\frac{1}{n}\sum_{t=1}^{n}\left(\mathbf{A}_t-\mathbf{A}_k(t/n)\right)\mathbf{X}_t\check{\mathbf{X}}_t^{(k)\top}\\&=\frac{1}{n}\sum_{t=1}^{n}\left(\mathbf{A}_t-\mathbf{A}_k(t/n)\right)\mathbf{X}_t\mathbf{X}_t^{(k)\top}-\left(\frac{1}{n}\sum_{t=1}^{n}\left(\mathbf{A}_t-\mathbf{A}_k(t/n)\right)\mathbf{X}_t\mathbf{Z}_t^{\top}\right)\left(\frac{1}{n}\sum_{t=1}^{n}\mathbf{Z}_t\mathbf{Z}_t^{\top}\right)^{-1}\left(\frac{1}{n}\sum_{t=1}^{n}\mathbf{Z}_t\mathbf{X}_t^{(k)\top}\right)\\&=O_p\left(\frac{n}{k^q}\right)+O_p\left(\frac{\sqrt{n}}{k^q}\right)=O_p\left(\frac{n}{k^q}\right)=o_p(1),\end{aligned}\tag{A.7}$$

where the second equality again follows from Lemma C.4(ii). What remains is to substitute (A.5)–(A.7) into (A.4). The conclusion then follows by applying the same arguments as those used in the proof of Theorem 1, so details are omitted.

A.5. Proof of Theorem 4

Let $\mathbf{Z}_t = \left(\Delta\mathbf{X}_{t-K}^\top, \cdots, \Delta\mathbf{X}_{t+K}^\top\right)^\top$. The steps follow similarly as in the proofs of Theorem 3. Estimation is based on the following regression

$$\check{\mathbf{y}}_t = \mathbf{B}^{(k)}\check{\mathbf{X}}_t^{(k)} + \check{\mathbf{u}}_{0t}^{(k)},$$

where $\check{\mathbf{u}}_{0t}^{(k)} = \mathbf{v}_t + \sum_{|j|>K}\mathbf{\Gamma}_j\mathbf{X}_{t-j} + (\mathbf{A}_t - \mathbf{A}_k(t/n))\,\mathbf{X}_t + \left(\mathbf{B}^{(k)}\widehat{\mathbf{\Gamma}}_{XZ} - \widehat{\mathbf{B}}_{YZ} + \mathbf{\Gamma}\right)\mathbf{Z}_t$. Then, we have

$$\begin{aligned}
\widehat{\mathbf{B}}_n^{(k)} - \mathbf{B}^{(k)} &= \left(\sum_{t=K+1}^{n-2K}\check{\mathbf{u}}_{0t}^{(k)}\check{\mathbf{X}}_t^{(k)\top}\right)\left(\sum_{t=K+1}^{n-2K}\check{\mathbf{X}}_t^{(k)}\check{\mathbf{X}}_t^{(k)\top}\right)^{-1} \\
&= \left(\sum_{t=K+1}^{n-2K}\widetilde{\mathbf{v}}_t\check{\mathbf{X}}_t^{(k)\top}\right)\left(\sum_{t=K+1}^{n-2K}\check{\mathbf{X}}_t^{(k)}\check{\mathbf{X}}_t^{(k)\top}\right)^{-1} \\
&\quad + \left(\sum_{t=K+1}^{n-2K}(\mathbf{A}_t - \mathbf{A}_k(t/n))\,\mathbf{X}_t\check{\mathbf{X}}_t^{(k)\top}\right)\left(\sum_{t=K+1}^{n-2K}\check{\mathbf{X}}_t^{(k)}\check{\mathbf{X}}_t^{(k)\top}\right)^{-1},
\end{aligned} \tag{A.8}$$

where $\widetilde{\mathbf{v}}_t = \mathbf{v}_t + \sum_{|j|>K}\mathbf{\Gamma}_j\mathbf{X}_{t-j}$.

First, note that

$$\begin{aligned}
&\frac{1}{(n-2K)^2}\sum_{t=K+1}^{n-2K}\check{\mathbf{X}}_t^{(k)}\check{\mathbf{X}}_t^{(k)\top} \\
=&\frac{1}{(n-2K)^2}\sum_{t=K+1}^{n-2K}\mathbf{X}_t^{(k)}\mathbf{X}_t^{(k)\top} \\
&-\left(\frac{1}{(n-2K)^2}\sum_{t=K+1}^{n-2K}\mathbf{X}_t^{(k)}\mathbf{Z}_t^\top\right)\left(\frac{1}{n-2K}\sum_{t=1}^{n}\mathbf{Z}_t\mathbf{Z}_t^\top\right)^{-1}\left(\frac{1}{n-2K}\sum_{t=K+1}^{n-2K}\mathbf{Z}_t\mathbf{X}_t^{(k)\top}\right) \\
=&\frac{1}{(n-2K)^2}\sum_{t=K+1}^{n-2K}\mathbf{X}_t^{(k)}\mathbf{X}_t^{(k)\top} \\
&-\left(\frac{1}{(n-2K)^2}\sum_{t=K+1}^{n-2K}\mathbf{X}_t^{(k)}\mathbf{Z}_t^\top\right)\left[\left(\frac{1}{n-2K}\sum_{t=1}^{n}\mathbf{Z}_t\mathbf{Z}_t^\top\right)^{-1} - \mathbf{\Sigma}_{zz}^{-1}\right]\left(\frac{1}{n-2K}\sum_{t=K+1}^{n-2K}\mathbf{Z}_t\mathbf{X}_t^{(k)\top}\right) \\
&-\left(\frac{1}{(n-2K)^2}\sum_{t=K+1}^{n-2K}\mathbf{X}_t^{(k)}\mathbf{Z}_t^\top\right)\mathbf{\Sigma}_{zz}^{-1}\left(\frac{1}{n-2K}\sum_{t=K+1}^{n-2K}\mathbf{Z}_t\mathbf{X}_t^{(k)\top}\right) \\
:=&\frac{1}{(n-2K)^2}\sum_{t=K+1}^{n-2K}\mathbf{X}_t^{(k)}\mathbf{X}_t^{(k)\top} + \mathbf{R}_{xx,1} + \mathbf{R}_{xx,2}.
\end{aligned}$$

By Lemma C.5, we have

$$\begin{aligned}
\|\mathbf{R}_{xx,1}\| &\leq \left\| \frac{1}{(n-2K)^2} \sum_{t=K+1}^{n-2K} \mathbf{X}_t^{(k)} \mathbf{Z}_t^\top \right\| \left\| \left(\frac{1}{n-2K} \sum_{t=1}^{n} \mathbf{Z}_t \mathbf{Z}_t^\top \right)^{-1} - \boldsymbol{\Sigma}_{zz}^{-1} \right\|_{sp} \left\| \frac{1}{n-2K} \sum_{t=K+1}^{n-2K} \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right\| \\
&= O_p\left(\frac{\sqrt{Kk}}{n} \right) O_p\left(\frac{K}{\sqrt{n}} \right) O_p\left(\sqrt{Kk} \right) = O_p\left(\frac{K^2 k}{n^{3/2}} \right),
\end{aligned}$$

and

$$\begin{aligned}
\|\mathbf{R}_{xx,2}\| &\leq \left\| \frac{1}{(n-2K)^2} \sum_{t=K+1}^{n-2K} \mathbf{X}_t^{(k)} \mathbf{Z}_t^\top \right\| \|\boldsymbol{\Sigma}_{zz}^{-1}\|_{sp} \left\| \frac{1}{n-2K} \sum_{t=K+1}^{n-2K} \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right\| \\
&= O_p\left(\frac{\sqrt{Kk}}{n} \right) O_p(1) O_p\left(\sqrt{Kk} \right) = O_p\left(\frac{Kk}{n} \right).
\end{aligned}$$

Given Assumption 3 and the condition $\frac{K^2}{n^{1-\gamma}} \to 0$, both terms above are just $o_p(1)$. This implies that

$$\begin{aligned}
\frac{1}{(n-2K)^2} \sum_{t=K+1}^{n-2K} \check{\mathbf{X}}_t^{(k)} \check{\mathbf{X}}_t^{(k)\top} &= \frac{1}{(n-2K)^2} \sum_{t=K+1}^{n-2K} \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} + o_p(1) \\
&= \frac{1}{n^2} \sum_{t=1}^{n} \mathbf{X}_t^{(k)} \mathbf{X}_t^{(k)\top} + o_p(1) = \boldsymbol{\Lambda}_{xx}^{(k)} + o_p(1),
\end{aligned}$$

provided that $K = o(n)$, where $\boldsymbol{\Lambda}_{xx}^{(k)}$ is defined as in Theorem 1.

Second, note that

$$
\begin{aligned}
&\frac{1}{n-2K}\sum_{t=K+1}^{n-K}\widetilde{\mathbf{v}}_t\check{\mathbf{X}}_t^{(k)\top}\\
=&\frac{1}{n-2K}\sum_{t=K+1}^{n-K}\widetilde{\mathbf{v}}_t\mathbf{X}_t^{(k)\top}-\left(\frac{1}{n-2K}\sum_{t=K+1}^{n-2K}\widetilde{\mathbf{v}}_t\mathbf{Z}_t^\top\right)\left(\frac{1}{n-2K}\sum_{t=1}^{n}\mathbf{Z}_t\mathbf{Z}_t^\top\right)^{-1}\left(\frac{1}{n-2K}\sum_{t=K+1}^{n-2K}\mathbf{Z}_t\mathbf{X}_t^{(k)\top}\right)\\
=&\frac{1}{n-2K}\sum_{t=K+1}^{n-K}\widetilde{\mathbf{v}}_t\mathbf{X}_t^{(k)\top}\\
&-\left(\frac{1}{n-2K}\sum_{t=K+1}^{n-2K}\widetilde{\mathbf{v}}_t\mathbf{Z}_t^\top\right)\left[\left(\frac{1}{n-2K}\sum_{t=1}^{n}\mathbf{Z}_t\mathbf{Z}_t^\top\right)^{-1}-\mathbf{\Sigma}_{zz}^{-1}\right]\left(\frac{1}{n-2K}\sum_{t=K+1}^{n-2K}\mathbf{Z}_t\mathbf{X}_t^{(k)\top}\right)\\
&-\left(\frac{1}{n-2K}\sum_{t=K+1}^{n-2K}\widetilde{\mathbf{v}}_t\mathbf{Z}_t^\top\right)\mathbf{\Sigma}_{zz}^{-1}\left(\frac{1}{n-2K}\sum_{t=K+1}^{n-2K}\mathbf{Z}_t\mathbf{X}_t^{(k)\top}\right)\\
:=&\frac{1}{n-2K}\sum_{t=K+1}^{n-K}\widetilde{\mathbf{v}}_t\mathbf{X}_t^{(k)\top}+\mathbf{R}_{vx,1}+\mathbf{R}_{vx,2}.
\end{aligned}
$$

It follows again from Lemma C.5 that

$$
\begin{aligned}
\|\mathbf{R}_{vx,1}\|\leq&\left\|\frac{1}{n-2K}\sum_{t=K+1}^{n-2K}\widetilde{\mathbf{v}}_t\mathbf{Z}_t^\top\right\|\left\|\left(\frac{1}{n-2K}\sum_{t=1}^{n}\mathbf{Z}_t\mathbf{Z}_t^\top\right)^{-1}-\mathbf{\Sigma}_{zz}^{-1}\right\|_{sp}\left\|\frac{1}{n-2K}\sum_{t=K+1}^{n-2K}\mathbf{Z}_t\mathbf{X}_t^{(k)\top}\right\|\\
&=O_p\left(\sqrt{\frac{K}{n}}\right)O_p\left(\frac{K}{\sqrt{n}}\right)O_p\left(\sqrt{Kk}\right)=O_p\left(\frac{K^2\sqrt{k}}{n}\right),
\end{aligned}
$$

and

$$
\begin{aligned}
\|\mathbf{R}_{vx,2}\|\leq&\left\|\frac{1}{n-2K}\sum_{t=K+1}^{n-2K}\widetilde{\mathbf{v}}_t\mathbf{Z}_t^\top\right\|\left\|\mathbf{\Sigma}_{zz}^{-1}\right\|_{sp}\left\|\frac{1}{n-2K}\sum_{t=K+1}^{n-2K}\mathbf{Z}_t\mathbf{X}_t^{(k)\top}\right\|\\
&=O_p\left(\sqrt{\frac{K}{n}}\right)O_p(1)O_p\left(\sqrt{Kk}\right)=O_p\left(\frac{K\sqrt{k}}{\sqrt{n}}\right).
\end{aligned}
$$

Again, given Assumption 3 and the condition $\frac{K^2}{n^{1-\gamma}}\to 0$, both terms above are just $o_p(1)$. This implies that

$$\begin{aligned}\frac{1}{n-2K}\sum_{t=K+1}^{n-K}\widetilde{\mathbf{v}}_t\check{\mathbf{X}}_t^{(k)\top} &= \frac{1}{n-2K}\sum_{t=K+1}^{n-K}\widetilde{\mathbf{v}}_t\mathbf{X}_t^{(k)\top}+o_p(1)\\ &= \frac{1}{n}\sum_{t=1}^{n}\mathbf{v}_t\mathbf{X}_t^{(k)\top}+o_p(1).\end{aligned}$$

Now, substituting all results back to (A.8) yields

$$n\cdot\left(\widehat{\mathbf{B}}_n^{(k)}-\mathbf{B}^{(k)}\right)=\left(\frac{1}{n}\sum_{t=1}^{n}\mathbf{v}_t\mathbf{X}_t^{(k)\top}\right)\left(\frac{1}{n^2}\sum_{t=1}^{n}\mathbf{X}_t^{(k)}\mathbf{X}_t^{(k)\top}\right)^{-1}+o_p(1).$$

Let $\tilde{\mathbf{\Gamma}}_{n,ux}^{(k)} := \frac{1}{n}\sum_{t=1}^{n}\mathbf{v}_t\mathbf{X}_t^{(k)\top}$ and

$$\widetilde{\mathbb{V}}_{n(k)}(\tau_t)=\left\{\mathbf{T}_{(k)}^{\top}(\tau_t)\left(\mathbf{\Lambda}_{n,xx}^{(k)}\right)^{-1}\mathbf{T}_{(k)}(\tau_t)\right\}\otimes\widehat{\mathbf{\Omega}}_{n,0\cdot x},$$

where $\widehat{\mathbf{\Omega}}_{n,0\cdot x}$ is a consistent estimator of $\mathbf{\Omega}_{0\cdot x}=\mathbf{\Omega}_{00}-\mathbf{\Omega}_{0x}\mathbf{\Omega}_{xx}^{-1}\mathbf{\Omega}_{x0}$. Then, we have

$$\begin{aligned}&n\left(\widetilde{\mathbb{V}}_{n(k)}(\tau_t)\right)^{-1/2}\operatorname{vec}\left(\widehat{\mathbf{A}}_n^{(k)}(\tau_t)-\mathbf{A}(\tau_t)+\mathbf{\Gamma}(\tau_t)\right)\\ &=\left(\left\{\mathbf{T}_{(k)}^{\top}(u)\left(\mathbf{\Lambda}_{n,xx}^{(k)}\right)^{-1}\mathbf{T}_{(k)}(u)\right\}\otimes\widehat{\mathbf{\Omega}}_{n,0\cdot x}\right)^{-1/2}\left(\mathbf{T}_{(k)}^{\top}(\tau_t)\otimes\mathbf{I}_d\right)\operatorname{vec}\left\{\tilde{\mathbf{\Gamma}}_{n,ux}^{(k)}\left(\mathbf{\Lambda}_{n,xx}^{(k)}\right)^{-1}\right\}+o_p(1).\end{aligned}$$

Following similar procedures as used in the proof of Corollary 1, we have the joint stochastic expansion

$$\begin{pmatrix}\mathbf{\Lambda}_{n,xx}^{(k)}\\ \tilde{\mathbf{\Gamma}}_{n,ux}^{(k)}\end{pmatrix}=\begin{pmatrix}\mathbf{\Lambda}_{xx}^{(k)}\\ \widetilde{\mathbf{\Gamma}}_{ux}^{(k)}\end{pmatrix}+o_p(1),$$

where

$$\widetilde{\mathbf{\Gamma}}_{ux}^{(k)}:=\int_0^1 d\mathbf{B}_{0.\cdot x}(r)\left(\left(\boldsymbol{\psi}_{(k)}(r)\otimes\mathbf{B}_r(\mathbf{\Omega}_{xx})\right)^{\top}\right)^{\top},$$

where $\mathbf{B}_{0.\cdot x}(r) := \mathbf{B}_v(r)=\mathbf{B}_0(r)-\mathbf{\Omega}_{0x}\mathbf{\Omega}_{xx}^{-1}\mathbf{B}_X(r)$. By construction, $\mathbf{B}_{0.\cdot x}(r)$ and $\mathbf{B}_X(r)$ are uncorrelated and hence independently distributed. Then, as $\widehat{\mathbf{\Omega}}_{n,0\cdot c}\xrightarrow{p}\mathbf{\Omega}_{0\cdot c}$ implies $\widetilde{\mathbb{V}}_{n(k)}(\tau_t)=\mathbb{V}_{(k)}(\tau_t)+o_p(1)$, by Slutsky's theorem and continuous mapping theorem, we obtain

$$n\left(\widetilde{\mathbb{V}}_{n(k)}(\tau_t)\right)^{-1/2}\operatorname{vec}\left(\widehat{\mathbf{A}}_n^{(k)}(\tau_t)-\mathbf{A}(\tau_t)+\mathbf{\Gamma}(\tau_t)\right)\xrightarrow{d}\mathcal{N}\left(\mathbf{0}_{dK_1\times 1},\mathbf{I}_{dK_1}\right).$$

B — Justification of the specification (24)

First, we show that the autocovariance matrix of $\mathbf{u}_t$: $\boldsymbol{\Gamma}_u(k) := \sum_{j=0}^{\infty} \boldsymbol{\Phi}_j \boldsymbol{\Sigma}_\varepsilon \boldsymbol{\Phi}_{j+k}^\top$, is absolute summable. Under Assumption 1, $\boldsymbol{\Sigma}_\varepsilon$ is positive definite and $\sum_{m=0}^{\infty} \|\boldsymbol{\Phi}_m\| < \infty$, we have

$$
\begin{aligned}
\sum_{k=0}^{\infty} \|\boldsymbol{\Gamma}_u(k)\| &\leq \sum_{k=0}^{\infty} \sum_{j=0}^{\infty} \|\boldsymbol{\Phi}_j\| \, \|\boldsymbol{\Sigma}_\varepsilon\| \, \left\|\boldsymbol{\Phi}_{j+k}^\top\right\| \\
&= \|\boldsymbol{\Sigma}_\varepsilon\| \sum_{j=0}^{\infty} \|\boldsymbol{\Phi}_j\| \left(\sum_{k=0}^{\infty} \left\|\boldsymbol{\Phi}_{j+k}^\top\right\| \right) \\
&\leq \|\boldsymbol{\Sigma}_\varepsilon\| \left(\sum_{m=0}^{\infty} \|\boldsymbol{\Phi}_m\| \right)^2 < \infty.
\end{aligned}
$$

Second, note that the spectral density matrix of $\mathbf{u}_t$ is given by $\mathbf{f}_{uu}(\lambda) = \frac{1}{2\pi} \boldsymbol{\Phi}\left(e^{-i\lambda}\right) \boldsymbol{\Sigma}_\varepsilon \boldsymbol{\Phi}^\top \left(e^{i\lambda}\right)$. Under Assumption 1, $\boldsymbol{\Sigma}_\varepsilon$ is positive definite and $\boldsymbol{\Phi}\left(e^{-i\lambda}\right)$ is nonsingular for all $\lambda \in [-\pi, \pi]$. Since $\mathbf{f}_{uu}(\lambda)$ is continuous in λ (implied by the absolute summability of $\boldsymbol{\Gamma}_u(k)$, $\forall k$) and $[-\pi, \pi]$ is compact, we have $\inf_{\lambda \in [-\pi,\pi]} \lambda_{\min}\left(f_{uu}(\lambda)\right) > 0$.

Given that $\sum_{k=0}^{\infty} \|\boldsymbol{\Gamma}_u(k)\| < \infty$ and $\inf_{\lambda \in [-\pi,\pi]} \lambda_{\min}\left(f_{uu}(\lambda)\right) > 0$, the conditions in Theorem 8.3.1 in Brillinger 1975 are satisfied. We may write

$$
\mathbf{u}_{0t} = \sum_{j=-\infty}^{\infty} \boldsymbol{\Gamma}_j \mathbf{u}_{xt} + \mathbf{v}_t, \tag{B.1}
$$

where $\boldsymbol{\Gamma}_j = \frac{1}{2\pi} \int_{-\pi}^{\pi} \boldsymbol{\Gamma}(\lambda) e^{ij\lambda} d\lambda$ satisfies $\sum_{j=-\infty}^{\infty} \|\boldsymbol{\Gamma}_j\| < \infty$. $\boldsymbol{\Gamma}(\lambda) = \mathbf{f}_{u_0 u_x}(\lambda) \mathbf{f}_{u_x u_x}^{-1}(\lambda)$ is the complex regression coefficient of $\mathbf{u}_{0t}$ on $\mathbf{u}_{xt}$ at frequency λ. The error series $\mathbf{v}_t$ has spectral density matrix

$$
\mathbf{f}_{vv}(\lambda) = \mathbf{f}_{u_0 u_0}(\lambda) - \mathbf{f}_{u_0 u_x}(\lambda) \mathbf{f}_{u_x u_x}^{-1}(\lambda) \mathbf{f}_{u_x u_0}(\lambda).
$$

This implies that (24) is achieved by truncating at $|j| = K$ so that

$$
\mathbf{y}_t = \mathbf{B}^{(k)} \mathbf{X}_t^{(k)} + \sum_{j=-K}^{K} \boldsymbol{\Gamma}_j \mathbf{X}_{t-j} + \widetilde{\mathbf{v}}_t^{(k)},
$$

where $\widetilde{\mathbf{v}}_t^{(k)} = \mathbf{v}_t + \sum_{|j|>K} \boldsymbol{\Gamma}_j \mathbf{X}_{t-j} + \left(\mathbf{A}_t - \mathbf{A}_k(\tau_t)\right) \mathbf{X}_t$.

Finally, since the cross spectral density matrix of $\mathbf{u}_{xt}$ and $\mathbf{v}_t$ is given by

$$\mathbf{f}_{vu_x}(\lambda) = \mathbf{f}_{u_0u_0}(\lambda) - \mathbf{\Gamma}(\lambda)\mathbf{f}_{u_xu_x}(\lambda) = \mathbf{0}_{d_1\times K_1}, \ \ \forall\lambda,$$

we have

$$\mathbb{E}\left[\mathbf{v}_t\mathbf{u}_{x,t-k}^\top\right] = \int_{-\pi}^{\pi}\mathbf{f}_{vu_x}(\lambda)e^{ik\lambda}\,d\lambda = \mathbf{0}_{d_1\times K_1}, \ \ \forall k.$$

In addition, since $\sum_{j=-\infty}^{\infty}\|\mathbf{\Gamma}_j\| < \infty$ and $(\mathbf{u}_{xt})_t$ is weakly stationary, the process $\left(\sum_{j=-\infty}^{\infty}\mathbf{\Gamma}_j\mathbf{u}_{xt}\right)_t$ is also weakly stationary, so does $(\mathbf{v}_t)_t$.

C — Useful lemmas

Lemma C.1. Under Assumptions 1-3, we have, as $n\to\infty$

$$\frac{1}{n^2}\sum_{t=1}^{n}\mathbf{X}_t^{(k)}\mathbf{X}_t^{\top(k)} = \mathbf{\Lambda}_{xx}^{(k)} + o_p(1),$$

where

$$\mathbf{\Lambda}_{xx}^{(k)} := \int_0^1\left\{\boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r)\otimes\left(\mathbf{B}_X(r)\mathbf{B}_X^\top(r)\right)\right\}dr$$

satisfies $\left\|\left(\mathbf{\Lambda}_{xx}^{(k)}\right)^{-1}\right\|_{sp} = O_p(1).$

Proof. Let $\mathbf{V}_{n,\lfloor nr\rfloor}^x = \mathbf{\Phi}_x(1)\frac{1}{\sqrt{n}}\sum_{s=1}^{\lfloor nr\rfloor}\boldsymbol{\varepsilon}_s$. Using the BN decomposition as in Phillips and Solo 1992, we have

$$\mathbf{X}_t - \mathbf{X}_{t-1} = \mathbf{u}_{xt} = \mathbf{\Phi}_x(1)\boldsymbol{\varepsilon}_t - \Delta\tilde{\mathbf{u}}_{xt},$$

where $\tilde{\mathbf{u}}_{xt} = \sum_{j=0}^{\infty}\widetilde{\mathbf{\Phi}}_{xj}\boldsymbol{\varepsilon}_{t-j}$ with $\widetilde{\mathbf{\Phi}}_{xj} = \sum_{k=j+1}^{\infty}\mathbf{\Phi}_{xj}$. Then,

$$\mathbf{X}_t = \mathbf{\Phi}_x(1)\sum_{s=1}^{t}\boldsymbol{\varepsilon}_s + \tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0 = \sqrt{n}\mathbf{V}_{n,t}^x + \tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0.$$

We have

$$
\begin{aligned}
&\frac{1}{n^2}\sum_{t=1}^{n}\mathbf{X}_t^{(k)}\mathbf{X}_t^{(k)\top}\\
&=\frac{1}{n^2}\sum_{t=1}^{n}\left(\boldsymbol{\psi}_{(k)}(t/n)\otimes\mathbf{X}_t\right)\left(\boldsymbol{\psi}_{(k)}(t/n)\otimes\mathbf{X}_t\right)^\top\\
&=\frac{1}{n^2}\sum_{t=1}^{n}\left(\boldsymbol{\psi}_{(k)}(t/n)\boldsymbol{\psi}_{(k)}^\top(t/n)\otimes\mathbf{X}_t\mathbf{X}_t^\top\right)\\
&=\frac{1}{n^2}\sum_{t=1}^{n}\left(\boldsymbol{\psi}_{(k)}(t/n)\boldsymbol{\psi}_{(k)}^\top(t/n)\otimes\left(\sqrt{n}\mathbf{V}_{n,t}^x\right)\left(\sqrt{n}\mathbf{V}_{n,t}^x\right)^\top\right)\\
&\quad+\frac{1}{n^2}\sum_{t=1}^{n}\left(\boldsymbol{\psi}_{(k)}(t/n)\boldsymbol{\psi}_{(k)}^\top(t/n)\otimes\left(\sqrt{n}\mathbf{V}_{n,t}^x\right)\left(\tilde{\mathbf{u}}_{x0}-\tilde{\mathbf{u}}_{xt}+\mathbf{X}_0\right)^\top\right)\\
&\quad+\frac{1}{n^2}\sum_{t=1}^{n}\left(\boldsymbol{\psi}_{(k)}^\top(t/n)\boldsymbol{\psi}_{(k)}^\top(t/n)\otimes\left(\tilde{\mathbf{u}}_{x0}-\tilde{\mathbf{u}}_{xt}+\mathbf{X}_0\right)\left(\sqrt{n}\mathbf{V}_{n,t}^x\right)^\top\right)\\
&\quad+\frac{1}{n^2}\sum_{t=1}^{n}\left(\boldsymbol{\psi}_{(k)}(t/n)\boldsymbol{\psi}_{(k)}^\top(t/n)\otimes\left(\tilde{\mathbf{u}}_{x0}-\tilde{\mathbf{u}}_{xt}+\mathbf{X}_0\right)\left(\tilde{\mathbf{u}}_{x0}-\tilde{\mathbf{u}}_{xt}+\mathbf{X}_0\right)^\top\right)\\
&:=\boldsymbol{\Lambda}_n(1)+\boldsymbol{\Lambda}_n(2)+\boldsymbol{\Lambda}_n(3)+\boldsymbol{\Lambda}_n(4),
\end{aligned}
$$

so that

$$
\left\|\frac{1}{n^2}\sum_{t=1}^{n}\mathbf{X}_t^{(k)}\mathbf{X}_t^{(k)\top}-\boldsymbol{\Lambda}_n(1)\right\|\leq\left\|\boldsymbol{\Lambda}_n(2)\right\|+\left\|\boldsymbol{\Lambda}_n(3)\right\|+\left\|\boldsymbol{\Lambda}_n(4)\right\|.
$$

We first show that $\|\boldsymbol{\Lambda}_n(2)\|=O_p\left(\frac{k}{n^a}\right)$, $\|\boldsymbol{\Lambda}_n(3)\|=O_p\left(\frac{k}{n^a}\right)$, and $\|\boldsymbol{\Lambda}_n(4)\|=O_p\left(\frac{k}{n^{2a}}\right)$, where $a=\frac{p-2}{2p}$. This would imply that

$$
\frac{1}{n^2}\sum_{t=1}^{n}\mathbf{X}_t^{(k)}\mathbf{X}_t^{(k)\top}=\boldsymbol{\Lambda}_n(1)+O_p\left(\frac{k}{n^a}\right)=\boldsymbol{\Lambda}_n(1)+o_p(1). \tag{C.1}
$$

Let us start with $\|\mathbf{\Lambda}_n(2)\|$. Notice that

$$\begin{aligned}\|\mathbf{\Lambda}_n(2)\| &\leq \frac{1}{n^2}\sum_{t=1}^{n}\left\|\boldsymbol{\psi}_{(k)}(t/n)\boldsymbol{\psi}_{(k)}^\top(t/n)\right\|\left\|(\sqrt{n}\,\mathbf{V}_{n,t}^x)(\tilde{\mathbf{u}}_{x0}-\tilde{\mathbf{u}}_{xt}+\mathbf{X}_0)^\top\right\| \\ &\leq \frac{1}{n^2}\sum_{t=1}^{n}\left\|\boldsymbol{\psi}_{(k)}(t/n)\boldsymbol{\psi}_{(k)}^\top(t/n)\right\|\sqrt{n}\,\left\|\mathbf{V}_{n,t}^x\right\|\|\tilde{\mathbf{u}}_{x0}-\tilde{\mathbf{u}}_{xt}+\mathbf{X}_0\| \\ &\leq \max_{1\leq t\leq n}\left\|\frac{1}{\sqrt{n}}(\tilde{\mathbf{u}}_{x0}-\tilde{\mathbf{u}}_{xt}+\mathbf{X}_0)\right\|\cdot\frac{1}{n^2}\sum_{t=1}^{n}\left\|\boldsymbol{\psi}_{(k)}(t/n)\boldsymbol{\psi}_{(k)}^\top(t/n)\right\|\cdot n\,\left\|\mathbf{V}_{n,t}^x\right\|.\end{aligned}$$

By standard Riemann–sum approximation for the integral, we have

$$\frac{1}{n}\sum_{t=1}^{n}\left\|\boldsymbol{\psi}_{(k)}(t/n)\boldsymbol{\psi}_{(k)}^\top(t/n)\right\| = \int_0^1\left\|\boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r)\right\|dr+O(1/n)=O(1)k.$$

In addition, by Lemma C.6, we have

$$\max_{1\leq t\leq n}\left\|\frac{1}{\sqrt{n}}(\tilde{\mathbf{u}}_{x0}-\tilde{\mathbf{u}}_{xt}+\mathbf{X}_0)\right\| = O_p\left(\frac{1}{n^a}\right),$$

for $a=\frac{p-2}{2p}$. Substituting all these results above yields

$$\|\mathbf{\Lambda}_n(2)\| = O_p\left(\frac{k}{n^a}\right).$$

Similar arguments lead to $\|\mathbf{\Lambda}_n(3)\| = O_p\left(\frac{k}{n^a}\right)$ and $\|\mathbf{\Lambda}_n(4)\| = O_p\left(\frac{k}{n^{2a}}\right)$.

We now move on to $\mathbf{\Lambda}_n(1)$. By construction, $\mathbf{V}_{n,0}=\mathbf{0}$. We have

$$\begin{aligned}&\frac{1}{n^2}\sum_{t=1}^{n}\left(\boldsymbol{\psi}_{(k)}(t/n)\boldsymbol{\psi}_{(k)}^\top(t/n)\otimes\left(\sqrt{n}\mathbf{V}_{n,t}^x\right)\left(\sqrt{n}\mathbf{V}_{n,t}^x\right)^\top\right) \\ &= \int_0^1\left(\boldsymbol{\psi}_{(k)}(\lfloor nr\rfloor/n)\boldsymbol{\psi}_{(k)}^\top(\lfloor nr\rfloor/n)\otimes\mathbf{V}_{n,\lfloor nr\rfloor}^x\mathbf{V}_{n,\lfloor nr\rfloor}^{x\top}\right)dr+\frac{1}{n}\left(\boldsymbol{\psi}_{(k)}(1)\boldsymbol{\psi}_{(k)}^\top(1)\otimes\mathbf{V}_{n,n}^x\mathbf{V}_{n,n}^{x\top}\right) \\ &= \int_0^1\left(\boldsymbol{\psi}_{(k)}(\lfloor nr\rfloor/n)\boldsymbol{\psi}_{(k)}^\top(\lfloor nr\rfloor/n)\otimes\mathbf{V}_{n,\lfloor nr\rfloor}^x\mathbf{V}_{n,\lfloor nr\rfloor}^{x\top}\right)dr+O_p(k/n),\end{aligned}$$

where $\frac{1}{n}\left(\boldsymbol{\psi}_{(k)}(1)\boldsymbol{\psi}_{(k)}^\top(1)\otimes\mathbf{V}_{n,n}^x\mathbf{V}_{n,n}^{x\top}\right)=O_p(k/n)=o_p(1)$, due to the fact that $\left\|\psi_{(k)}(1)\psi_{(k)}(1)^\top\right\|=$

$O(1)\, k$ and (4). Notice that

$$
\begin{aligned}
&\int_0^1 \left(\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor/n)\boldsymbol{\psi}_{(k)}^\top(\lfloor nr \rfloor/n) - \boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r)\right) dr \\
= &\int_0^1 \left(\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor/n) - \boldsymbol{\psi}_{(k)}(r)\right) \boldsymbol{\psi}_{(k)}^\top(r)\, dr + \int_0^1 \boldsymbol{\psi}_{(k)}(r) \left(\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor/n) - \boldsymbol{\psi}_{(k)}(r)\right)^\top dr \\
&\quad + \int_0^1 \left(\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor/n) - \boldsymbol{\psi}_{(k)}(r)\right) \left(\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor/n) - \boldsymbol{\psi}_{(k)}(r)\right)^\top dr.
\end{aligned}
$$

Since

$$
\begin{aligned}
\left\|\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor/n) - \boldsymbol{\psi}_{(k)}(r)\right\|^2 &= 2\sum_{j=1}^{k-1} \left(\cos\left(j\pi \frac{\lfloor nr \rfloor}{n}\right) - \cos(j\pi r)\right)^2 \\
&\leq O(1) \left|\frac{\lfloor nr \rfloor}{n} - r\right|^2 \left(\sum_{j=1}^{k-1} j^2\right) \leq O(1) k^3/n^2,
\end{aligned}
$$

we have

$$
\left\|\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor/n) - \boldsymbol{\psi}_{(k)}(r)\right\| = O\left(\frac{k^{3/2}}{n}\right) = o(1).
$$

As by Assumption 3, $k = o(n^{2/3})$, this implies that

$$
\int_0^1 \left(\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor/n)\boldsymbol{\psi}_{(k)}^\top(\lfloor nr \rfloor/n) - \boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r)\right) dr = o(1). \tag{C.2}
$$

Thus,

$$
\begin{aligned}
\boldsymbol{\Lambda}_n(1) = &\int_0^1 \left(\boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor/n)\boldsymbol{\psi}_{(k)}^\top(\lfloor nr \rfloor/n) - \boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r)\right) \otimes \left(\mathbf{V}_{n,\lfloor nr \rfloor}^x \mathbf{V}_{n,\lfloor nr \rfloor}^{x\top}\right)\, dr \\
&\quad + \int_0^1 \left(\boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r) \otimes \mathbf{V}_{n,\lfloor nr \rfloor}^x \mathbf{V}_{n,\lfloor nr \rfloor}^{x\top}\right)\, dr \\
= &\int_0^1 \left(\boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r) \otimes \mathbf{V}_{n,\lfloor nr \rfloor}^x \mathbf{V}_{n,\lfloor nr \rfloor}^{x\top}\right)\, dr + o_p(1).
\end{aligned} \tag{C.3}
$$

By combining (C.1) and (C.3), we conclude that

$$
\frac{1}{n^2}\sum_{t=1}^n \mathbf{X}_t^{(k)}\mathbf{X}_t^{\top(k)} = \int_0^1 \left(\boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r) \otimes \mathbf{V}_{n,\lfloor nr \rfloor}^x \mathbf{V}_{n,\lfloor nr \rfloor}^{x\top}\right) dr + O_p\left(\frac{k}{n^a}\right).
$$

Finally, we show that

$$\left\| \int_0^1 \left(\boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r) \otimes \mathbf{V}_{n,\lfloor nr\rfloor}^{x}\mathbf{V}_{n,\lfloor nr\rfloor}^{x\top}\right) dr - \int_0^1 \left(\boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r) \otimes \mathbf{B}_X(r)\mathbf{B}_X^\top(r)\right) dr \right\|_{sp} = o_p(1). \tag{C.4}$$

For any unit vector $\mathbf{z} \in \mathbb{R}^{kK_1}$, where $\mathbf{z} = \mathrm{vec}(\mathbf{Z})$, we note that

$$\begin{aligned}
&\left| \mathbf{z}^\top \left(\int_0^1 \left\{ \boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r) \otimes \left(\mathbf{V}_{n,\lfloor nr\rfloor}^{x}\mathbf{V}_{n,\lfloor nr\rfloor}^{x\top} - \mathbf{B}_X(r)\mathbf{B}_X^\top(r)\right)\right\} dr \right) \mathbf{z} \right| \\
&= \left\| \int_0^1 \left(\mathbf{Z}\boldsymbol{\psi}_{(k)}(r)\right)^\top \left(\mathbf{V}_{n,\lfloor nr\rfloor}^{x}\mathbf{V}_{n,\lfloor nr\rfloor}^{x\top} - \mathbf{B}_X(r)\mathbf{B}_X^\top(r)\right) \left(\mathbf{Z}\boldsymbol{\psi}_{(k)}(r)\right) dr \right\| \\
&\leq \sup_{r\in[0,1]} \left\| \mathbf{V}_{n,\lfloor nr\rfloor}^{x}\mathbf{V}_{n,\lfloor nr\rfloor}^{x\top} - \mathbf{B}_X(r)\mathbf{B}_X^\top(r) \right\| \int_0^1 \left\| \mathbf{Z}\boldsymbol{\psi}_{(k)}(r) \right\|^2 dr \\
&\leq \sup_{r\in[0,1]} \left\| \mathbf{V}_{n,\lfloor nr\rfloor}^{x}\mathbf{V}_{n,\lfloor nr\rfloor}^{x\top} - \mathbf{B}_X(r)\mathbf{B}_X^\top(r) \right\| \mathrm{tr}\left(\mathbf{Z}^\top\mathbf{Z} \int_0^1 \boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r) dr \right) \\
&\leq \sup_{r\in[0,1]} \left\| \mathbf{V}_{n,\lfloor nr\rfloor}^{x}\mathbf{V}_{n,\lfloor nr\rfloor}^{x\top} - \mathbf{B}_X(r)\mathbf{B}_X^\top(r) \right\|,
\end{aligned}$$

where the last inequality follows from the fact that

$$\mathrm{tr}\left(\mathbf{Z}^\top\mathbf{Z} \int_0^1 \boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r) dr \right) = \mathrm{tr}\left(\mathbf{Z}^\top\mathbf{Z}\right) = \|\mathbf{Z}\|^2 = 1.$$

In addition, by (4) and the fact that $\mathbf{B}_X(\cdot)$ has continuous sample paths $a.s.$, we have

$$\sup_{r\in[0,1]} \left\| \mathbf{V}_{n,\lfloor nr\rfloor}^{x}\mathbf{V}_{n,\lfloor nr\rfloor}^{x\top} - \mathbf{B}_X(r)\mathbf{B}_X^\top(r) \right\| = o_p(1).$$

We now show that $\left\| \left(\boldsymbol{\Lambda}_{xx}^{(k)}\right)^{-1} \right\|_{sp} = O_p(1)$. For any fixed vector $\mathbf{z} \neq \mathbf{0}_{kK_1\times 1}$, we need to show that

$$\mathbf{z}^\top \left(\int_0^1 \left(\boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r)\right) \otimes \left(\mathbf{B}_X(r)\mathbf{B}_X^\top(r)\right) dr \right) \mathbf{z} > 0 \quad a.s.$$

Let $\mathrm{vec}(\mathbf{Z}) = \mathbf{z}$. Using the fact that

$$\left(\boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r)\right) \otimes \left(\mathbf{B}_X(r)\mathbf{B}_X^\top(r)\right) = \left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r)\right)\left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r)\right)^\top$$

and the identity

$$\left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r)\right)^\top \operatorname{vec}(\mathbf{Z}) = \operatorname{vec}\left(\mathbf{B}_X^\top(r)\mathbf{Z}\boldsymbol{\psi}_{(k)}(r)\right),$$

we obtain

$$\begin{aligned}
&\mathbf{z}^\top \left(\int_0^1 \left(\boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r)\right) \otimes \left(\mathbf{B}_X(r)\mathbf{B}_X^\top(r)\right) dr \right) \mathbf{z} \\
&= \int_0^1 \mathbf{z}^\top \left(\left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r)\right)\left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r)\right)^\top \right) \mathbf{z}\, dr \\
&= \int_0^1 \left(\left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r)\right)^\top \operatorname{vec}(\mathbf{Z}) \right)^\top \left(\left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r)\right)^\top \operatorname{vec}(\mathbf{Z}) \right) dr \\
&= \int_0^1 \left(\operatorname{vec}\left(\mathbf{B}_X^\top(r)\mathbf{Z}\boldsymbol{\psi}_{(k)}(r)\right)\right)^\top \left(\operatorname{vec}\left(\mathbf{B}_X^\top(r)\mathbf{Z}\boldsymbol{\psi}_{(k)}(r)\right)\right) dr \\
&= \int_0^1 \left(\mathbf{B}_X^\top(r)\mathbf{Z}\boldsymbol{\psi}_{(k)}(r)\right)^2 dr \;\geq\; 0 \quad a.s.
\end{aligned}$$

Note that $\mathbf{Z}\boldsymbol{\psi}_{(k)}(r)$ is $K_1 \times 1$, so $\mathbf{B}_X^\top(r)\mathbf{Z}\boldsymbol{\psi}_{(k)}(r)$ is a one-dimensional Brownian motion obtained as a linear combination of the components of $\mathbf{B}_X(r)$ whenever $\mathbf{Z}\boldsymbol{\psi}_{(k)}(r) \neq 0$. Suppose, for contradiction, that

$$\int_0^1 \left(\mathbf{B}_X^\top(r)\mathbf{Z}\boldsymbol{\psi}_{(k)}(r)\right)^2 dr = 0$$

holds on an event of positive probability. Then, on that event, the continuous process

$$W(r) := \mathbf{B}_X^\top(r)\mathbf{Z}\boldsymbol{\psi}_{(k)}(r)$$

satisfies $W(r) = 0$, except on a set of Lebesgue measure zero on $[0,1]$. Since $\boldsymbol{\Omega}_{xx}$ is full and the Brownian motion has continuous sample paths, this can only happen with positive probability if $\mathbf{Z}\boldsymbol{\psi}_{(k)}(r) = 0$ for almost every $r \in [0,1]$. If there exists $r_0 \in [0,1]$ such that $\mathbf{v} := \mathbf{Z}\boldsymbol{\psi}_{(k)}(r_0) \neq 0$, we have $\mathbf{v}^\top\boldsymbol{\Omega}_{xx}\mathbf{v} > 0$. Then, at the fixed r_0, $W(r_0) := \mathbf{B}_X(r_0)^\top\mathbf{v}$ has a Gaussian distribution with variance $r_0\mathbf{v}^\top\boldsymbol{\Omega}_{xx}\mathbf{v} > 0$, so that $\mathbb{P}\left(W(r_0) = 0\right) = 0$, which implies that $\mathbb{P}\left(|W(r_0)| > 0\right) = 1$. By continuity of $r \mapsto W(r)$, there exists a neighbourhood of r_0 such that $|W(r)| > 0$ for all $r \in \mathcal{N}_{r_0}$, which implies that $\int_0^1 W(r)^2 dr \geq \int_{\mathcal{N}_{r_0}} W(r)^2 dr > 0$, $a.s.$, a contradiction. Hence

$$\int_0^1 \left(\mathbf{Z}\boldsymbol{\psi}_{(k)}(r)\right)\left(\mathbf{Z}\boldsymbol{\psi}_{(k)}(r)\right)^\top dr = \mathbf{0}_{K_1 \times K_1}.$$

Since the basis is orthonormal, we have

$$\int_0^1 \boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r)\,dr = \mathbf{I}_k.$$

Therefore,

$$\int_0^1 \left(\mathbf{Z}\boldsymbol{\psi}_{(k)}(r)\right)\left(\mathbf{Z}\boldsymbol{\psi}_{(k)}(r)\right)^\top dr = \mathbf{Z}\left(\int_0^1 \boldsymbol{\psi}_{(k)}(r)\boldsymbol{\psi}_{(k)}^\top(r)\,dr\right)\mathbf{Z}^\top = \mathbf{Z}\mathbf{Z}^\top.$$

If the left-hand side equals $\mathbf{0}_{K_1\times K_1}$, then $\mathbf{Z}\mathbf{Z}^\top = \mathbf{0}_{K_1\times K_1}$, which implies $\mathbf{Z} = \mathbf{0}_{K_1\times k}$ and hence $\mathrm{vec}(\mathbf{Z}) = \mathbf{z} = \mathbf{0}_{kK_1\times 1}$, a contradiction. □

Lemma C.2. Under Assumptions 1-3, we have, as $n\to\infty$

$$\frac{1}{n}\sum_{t=1}^n \mathbf{u}_{0t}\mathbf{X}_t^{\top(k)} = \boldsymbol{\Gamma}_{ux}^{(k)} + o_p(1),$$

where

$$\boldsymbol{\Gamma}_{ux}^{(k)} := \int_0^1 d\mathbf{B}_0(r)\left(\boldsymbol{\psi}_{(k)}(r)\otimes\mathbf{B}_X(r)\right)^\top + \boldsymbol{\ell}_{1(k)}\otimes(\boldsymbol{\Lambda}_{0x}+\boldsymbol{\Sigma}_{0x})$$

satisfies $\left\|\boldsymbol{\Gamma}_{ux}^{(k)}\right\|_{sp} = O_p(1)$. $\boldsymbol{\ell}_{1(k)}$ denotes an $k\times 1$ vector with the first element equal to one and all remaining elements equal to zero.

Proof. Let $\mathbf{U}_n(r) = \boldsymbol{\Phi}_0(1)\frac{1}{\sqrt{n}}\sum_{s=1}^{\lfloor nr\rfloor}\boldsymbol{\varepsilon}_s$. We note that

$$\begin{aligned}\frac{1}{n}\sum_{t=1}^n \mathbf{u}_{0t}\mathbf{X}_t^{\top(k)} &= \frac{1}{n}\sum_{t=1}^n \mathbf{u}_{0t}\left(\boldsymbol{\psi}_{(k)}(t/n)\otimes\mathbf{X}_{t-1}\right)^\top + \frac{1}{n}\sum_{t=1}^n \mathbf{u}_{0t}\left(\boldsymbol{\psi}_{(k)}(t/n)\otimes\mathbf{u}_{xt}\right)^\top\\ &:= \Gamma_n(1)+\Gamma_n(2).\end{aligned}$$

We will show that

$$\Gamma_n(1) = \int_0^1 d\mathbf{U}_n(r)\,\left(\boldsymbol{\psi}_{(k)}(r)\otimes\mathbf{V}_{n,\lfloor nr\rfloor}^x\right)^\top + \boldsymbol{\ell}_{1(k)}^\top\otimes\boldsymbol{\Lambda}_{0x} + o_p(1), \tag{C.5}$$

$$\Gamma_n(2) = \boldsymbol{\ell}_{1(k)}^\top\otimes\boldsymbol{\Sigma}_{0x} + o_p(1). \tag{C.6}$$

These together with triangular inequality implies

$$\left\|\frac{1}{n}\sum_{t=1}^{n}\mathbf{u}_{0t}\mathbf{X}_t^{\top(k)}-\int_0^1 d\mathbf{U}_n(r)\ \left(\boldsymbol{\psi}_{(k)}(r)\otimes\mathbf{V}^x_{n,\lfloor nr\rfloor}\right)^\top-\boldsymbol{\ell}^\top_{1(k)}\otimes(\boldsymbol{\Lambda}_{0x}+\boldsymbol{\Sigma}_{0x})\right\|=o_p(1).$$

Let us start with (C.5). BN decomposition gives

$$\begin{aligned}
\Gamma_n(1)=&\boldsymbol{\Phi}_0(1)\frac{1}{n}\sum_{t=1}^{n}\boldsymbol{\varepsilon}_t\left(\boldsymbol{\psi}_{(k)}(t/n)\otimes\mathbf{X}_{t-1}\right)^\top-\frac{1}{n}\sum_{t=1}^{n}\Delta\tilde{\mathbf{u}}_{0t}\left(\boldsymbol{\psi}_{(k)}(t/n)\otimes\mathbf{X}_{t-1}\right)^\top\\
&=\boldsymbol{\Phi}_0(1)\frac{1}{\sqrt{n}}\sum_{t=1}^{n}\boldsymbol{\varepsilon}_t\left(\boldsymbol{\psi}_{(k)}(t/n)\otimes\mathbf{V}^x_{n,t-1}\right)^\top+\boldsymbol{\Phi}_0(1)\frac{1}{n}\sum_{t=1}^{n}\boldsymbol{\varepsilon}_t\left(\boldsymbol{\psi}_{(k)}(t/n)\otimes(\tilde{\mathbf{u}}_{x,0}-\tilde{\mathbf{u}}_{x,t-1}+\mathbf{X}_0)\right)^\top\\
&\quad-\frac{1}{n}\sum_{t=1}^{n}\Delta\tilde{\mathbf{u}}_{0t}\left(\boldsymbol{\psi}_{(k)}(t/n)\otimes\mathbf{X}_{t-1}\right)^\top\\
:=&\Gamma_n(1,1)+\Gamma_n(1,2)+\Gamma_n(1,3).
\end{aligned}$$

Now let us consider $\Gamma_n(1,1)$. We have

$$\begin{aligned}
\Gamma_n(1,1)&=\sum_{t=1}^{n}\left(\boldsymbol{\Phi}_0(1)\frac{1}{\sqrt{n}}\boldsymbol{\varepsilon}_t\right)\left(\boldsymbol{\psi}_{(k)}(t/n)\otimes\mathbf{V}^x_{n,t-1}\right)^\top\\
&=\int_0^1 d\mathbf{U}_n(r)\ \left(\boldsymbol{\psi}_{(k)}(\lfloor nr\rfloor/n)\otimes\mathbf{V}^x_{n,\lfloor nr\rfloor-1}\right)^\top\\
&=\int_0^1 d\mathbf{U}_n(r)\ \left(\boldsymbol{\psi}_{(k)}(r)\otimes\mathbf{V}^x_{n,\lfloor nr\rfloor}\right)^\top+\int_0^1 d\mathbf{U}_n(r)\ \left(\boldsymbol{\psi}_{(k)}(\lfloor nr\rfloor/n)\otimes\mathbf{V}^x_{n,\lfloor nr\rfloor-1}-\boldsymbol{\psi}_{(k)}(r)\otimes\mathbf{V}^x_{n,\lfloor nr\rfloor}\right)^\top\\
&=\int_0^1 d\mathbf{U}_n(r)\ \left(\boldsymbol{\psi}_{(k)}(r)\otimes\mathbf{V}^x_{n,\lfloor nr\rfloor}\right)^\top+o_p(1),
\end{aligned}$$

if we could show that

$$\int_0^1 d\mathbf{U}_n(r)\ \left(\boldsymbol{\psi}_{(k)}(\lfloor nr\rfloor/n)\otimes\mathbf{V}^x_{n,\lfloor nr\rfloor-1}-\boldsymbol{\psi}_{(k)}(r)\otimes\mathbf{V}^x_{n,\lfloor nr\rfloor}\right)^\top=o_p(1).\tag{C.7}$$

Note that

$$
\begin{aligned}
&\int_0^1 d\mathbf{U}_n(r)\ \left(\boldsymbol{\psi}_{(k)}(\lfloor nr\rfloor/n)\otimes\mathbf{V}^x_{n,\lfloor nr\rfloor-1}-\boldsymbol{\psi}_{(k)}(r)\otimes\mathbf{V}^x_{n,\lfloor nr\rfloor}\right)^\top\\
=&\int_0^1 d\mathbf{U}_n(r)\ \left\{\left(\boldsymbol{\psi}_{(k)}(\lfloor nr\rfloor/n)-\boldsymbol{\psi}_{(k)}(r)\right)\otimes\left(\mathbf{V}^x_{n,\lfloor nr\rfloor-1}-\mathbf{V}^x_{n,\lfloor nr\rfloor}\right)\right\}^\top\\
&+\int_0^1 d\mathbf{U}_n(r)\ \left\{\left(\boldsymbol{\psi}_{(k)}(\lfloor nr\rfloor/n)-\boldsymbol{\psi}_{(k)}(r)\right)\otimes\mathbf{V}^x_{n,\lfloor nr\rfloor}\right\}^\top\\
&+\int_0^1 d\mathbf{U}_n(r)\ \left\{\boldsymbol{\psi}_{(k)}(r)\otimes\left(\mathbf{V}^x_{n,\lfloor nr\rfloor-1}-\mathbf{V}^x_{n,\lfloor nr\rfloor}\right)\right\}^\top.
\end{aligned}
$$

We have shown in (C.2) that $\left\|\boldsymbol{\psi}_{(k)}(\lfloor nr\rfloor/n)-\boldsymbol{\psi}_{(k)}(r)\right\|=O\left(\frac{k^{3/2}}{n}\right)=o(1)$ for all $r\in[0,1]$. In addition, by the definition of $\mathbf{V}_{n,\lfloor nr\rfloor}$, $\left\|\mathbf{V}^x_{n,\lfloor nr\rfloor-1}-\mathbf{V}^x_{n,\lfloor nr\rfloor}\right\|=O_p(1/\sqrt{n})$. Then, each of the integrals above is $o_p(1)$ and hence (C.7) holds.

Moving on to $\Gamma_n(1,2)$. We note that

$$
\begin{aligned}
&\left\|\frac{1}{n}\sum_{t=1}^n\boldsymbol{\varepsilon}_t\left(\boldsymbol{\psi}_{(k)}(t/n)\otimes(\tilde{\mathbf{u}}_{x,0}-\tilde{\mathbf{u}}_{x,t-1}+\mathbf{X}_0)\right)^\top\right\|\\
\le&\ \frac{1}{n}\sum_{t=1}^n\left\|\boldsymbol{\varepsilon}_t\left(\boldsymbol{\psi}_{(k)}(t/n)\otimes(\tilde{\mathbf{u}}_{x,0}-\tilde{\mathbf{u}}_{x,t-1}+\mathbf{X}_0)\right)^\top\right\|\\
\le&\ \max_{1\le t\le n}\left\|\frac{1}{\sqrt{n}}\left(\tilde{\mathbf{u}}_{x0}-\tilde{\mathbf{u}}_{xt}+\mathbf{X}_0\right)\right\|\ \frac{1}{\sqrt{n}}\sum_{t=1}^n\|\boldsymbol{\varepsilon}_t\|\left\|\boldsymbol{\psi}_{(k)}(t/n)\right\|=O_p\left(\frac{\sqrt{k}}{n^a}\right),
\end{aligned}
$$

which follows from Lemma C.6 and the fact that $\left\|\boldsymbol{\psi}_{(k)}(t/n)\right\|=O(1)\sqrt{k}$, which holds uniformly over $t=1,2,\cdots,n$. This implies that $\|\Gamma_n(1,2)\|=O_p\left(\frac{\sqrt{k}}{n^a}\right)=o_p(1)$.

Finally, we consider $\Gamma_n(1,3)$. Summation by parts leads to

$$
\begin{aligned}
\Gamma_n(1,3)&=-\frac{1}{n}\sum_{t=1}^n\Delta\tilde{\mathbf{u}}_{0t}\left(\boldsymbol{\psi}_{(k)}(t/n)\otimes\mathbf{X}_{t-1}\right)^\top\\
&=\frac{1}{n}\sum_{t=1}^n\tilde{\mathbf{u}}_{0t}\left\{\left(\boldsymbol{\psi}_{(k)}((t+1)/n)\otimes\mathbf{X}_t\right)-\left(\boldsymbol{\psi}_{(k)}(t/n)\otimes\mathbf{X}_{t-1}\right)\right\}^\top+o_p(1)\\
&=\frac{1}{n}\sum_{t=1}^n\tilde{\mathbf{u}}_{0t}\left\{\left(\boldsymbol{\psi}_{(k)}((t+1)/n)-\boldsymbol{\psi}_{(k)}(t/n)\right)\otimes\mathbf{X}_t+\boldsymbol{\psi}_{(k)}(t/n)\otimes(\mathbf{X}_t-\mathbf{X}_{t-1})\right\}^\top+o_p(1)\\
&=\frac{1}{n}\sum_{t=1}^n\tilde{\mathbf{u}}_{0t}\left(\boldsymbol{\psi}_{(k)}(t/n)\otimes\mathbf{u}_{xt}\right)^\top+o_p(1),
\end{aligned}
$$

where the final equality follows from the Lipschitz-continuity of the base function, which implies that $\left\|\boldsymbol{\psi}_{(k)}((t+1)/n) - \boldsymbol{\psi}_{(k)}(t/n)\right\| = O(k^{3/2}/n)$, together with Cauchy–Schwarz inequality and the fact that $\frac{1}{n}\sum_{t=1}^{n}\|\mathbf{X}_t\|^2 = O_p(n)$, so that $\left\|\frac{1}{n}\sum_{t=1}^{n}\tilde{\mathbf{u}}_{0t}\left\{\left(\boldsymbol{\psi}_{(k)}((t+1)/n) - \boldsymbol{\psi}_{(k)}(t/n)\right)\otimes\mathbf{X}_t\right\}^\top\right\| = O_p\left(\frac{k^{3/2}}{\sqrt{n}}\right) = o_p(1)$. In addition,

$$
\begin{aligned}
&\frac{1}{n}\sum_{t=1}^{n}\tilde{\mathbf{u}}_{0t}\left(\boldsymbol{\psi}_{(k)}(t/n)\otimes\mathbf{u}_{xt}\right)^\top \\
&= \frac{1}{n}\sum_{t=1}^{n}\left(\boldsymbol{\psi}_{(k)}^\top(t/n)\otimes\left(\tilde{\mathbf{u}}_{0t}\mathbf{u}_{x,t}^\top - \boldsymbol{\Lambda}_{0x}\right)\right) + \left(\frac{1}{n}\sum_{t=1}^{n}\boldsymbol{\psi}_{(k)}^\top(t/n)\right)^\top \otimes\boldsymbol{\Lambda}_{0x} \\
&= \left(\int_0^1\boldsymbol{\psi}_{(k)}(r)\,dr\right)^\top\otimes\boldsymbol{\Lambda}_{0x} + O_p\left(\sqrt{\frac{k}{n}}\right) \\
&= \boldsymbol{\ell}_{1(k)}^\top\otimes\boldsymbol{\Lambda}_{0x} + o_p(1).
\end{aligned}
$$

The $O_p\left(\sqrt{k/n}\right)$ term follows by applying C.L.T. for the stationary and ergodic process $\left(\tilde{\mathbf{u}}_{0t}\mathbf{u}_{x,t}^\top - \boldsymbol{\Lambda}_{0x}\right)_t$, together with $\|\boldsymbol{\psi}_{(k)}(t/n)\| = O(1)\sqrt{k}$. Moreover, by the Riemann-sum approximation, $\frac{1}{n}\sum_{t=1}^{n}\boldsymbol{\psi}_{(k)}(t/n)\to\int_0^1\boldsymbol{\psi}_{(k)}(r)\,dr = \boldsymbol{\ell}_{1(k)}$ as $n\to\infty$, where $\boldsymbol{\ell}_{1(k)}$ is a unit vector with 1 in the first coordinate and 0 in all other coordinates.

By combining all results obtained above, we get

$$
\left\|\Gamma_n(1) - \int_0^1 d\mathbf{U}_n(r)\ \left(\boldsymbol{\psi}_{(k)}(r)\otimes\mathbf{V}_{n,\lfloor nr\rfloor}^x\right)^\top - \boldsymbol{\ell}_{1(k)}^\top\otimes\boldsymbol{\Lambda}_{0x}\right\| = o_p(1).
$$

Using Theorem 2.2 in Kurtz and Protter 1991, the joint convergence result of

$$
\begin{pmatrix}\frac{1}{\sqrt{n}}\sum_{t=1}^{\lfloor nr\rfloor}\mathbf{u}_{0t}\\[2ex] \frac{1}{\sqrt{n}}\sum_{t=1}^{\lfloor nr\rfloor}\mathbf{u}_{xt}\end{pmatrix} \Rightarrow \begin{pmatrix}\mathbf{B}_0(r)\\[2ex] \mathbf{B}_X(r)\end{pmatrix},
$$

on $D\big([0,1],\,\mathbb{R}^{d+K_1}\big)$, together with similar arguments as used to establish (C.4), we obtain

$$
\left\|\int_0^1 d\mathbf{U}_n(r)\ \left(\boldsymbol{\psi}_{(k)}(r)\otimes\mathbf{V}_{n,\lfloor nr\rfloor}^x\right)^\top - \int_0^1 d\mathbf{B}_0(r)\left(\boldsymbol{\psi}_{(k)}(r)\otimes\mathbf{B}_X(r)\right)^\top\right\|_{sp} = o_p(1).
$$

Turning to $\Gamma_n(2)$. This is straightforward, since same reasoning for $\frac{1}{n}\sum_{t=1}^{n}\tilde{\mathbf{u}}_{0t}\left(\boldsymbol{\psi}_{(k)}(t/n)\otimes\mathbf{u}_{xt}\right)^\top =$

$\boldsymbol{\ell}_{1(k)}^{\top} \otimes \boldsymbol{\Lambda}_{0x} + o_p(1)$ gives $\Gamma_n(2) = \boldsymbol{\ell}_{1(k)}^{\top} \otimes \boldsymbol{\Sigma}_{0x} + o_p(1)$. Thus, we have

$$\frac{1}{n}\sum_{t=1}^{n} \mathbf{u}_{0t}\mathbf{X}_t^{\top(k)} = \int_0^1 d\mathbf{B}_0(r)\left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r)\right)^{\top} + \boldsymbol{\ell}_{1(k)} \otimes (\boldsymbol{\Lambda}_{0x} + \boldsymbol{\Sigma}_{0x}) + o_p(1).$$

To show $\left\|\boldsymbol{\Gamma}_{ux}^{(k)}\right\|_{sp} = O_p(1)$, it is suffice to show that $\left\|\int_0^1 d\mathbf{B}_0(r)\left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r)\right)^{\top}\right\|_{sp} = O_p(1)$. Let $\mathbf{u} \in \mathbb{R}^d$ and $\mathbf{v} = \left(\mathbf{v}_0^{\top}, \cdots, \mathbf{v}_{k-1}^{\top}\right)^{\top} \in \mathbb{R}^{K_1 k}$ be unit vectors such that $\|\mathbf{u}\| = 1$ and $\|\mathbf{v}\| = 1$. Furthermore, define

$$q_{(k)} := \mathbf{u}^{\top}\left(\int_0^1 d\mathbf{B}_0(r)\left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_x(r)\right)^{\top}\right)\mathbf{v} := \int_0^1 g_{(k)}(r)\mathbf{u}^{\top} d\mathbf{B}_0(r).$$

By Itô isometry, we have

$$\mathbb{E}\left[q_{(k)}^2\right] = \left(\mathbf{u}^{\top}\boldsymbol{\Omega}_{00}\mathbf{u}\right)\mathbb{E}\left[\int_0^1 g_{(k)}^2(r)dr\right].$$

Recall that

$$g_{(k)}(r) = \left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_x(r)\right)^{\top}\mathbf{v} = \sum_{i=0}^{k-1} \psi_i(r)\mathbf{B}_x^{\top}(r)\mathbf{v}_i.$$

We then have

$$\begin{aligned}
\mathbb{E}\left[\int_0^1 g_{(k)}^2(r)dr\right] &= \int_0^1 \mathbb{E}\left[\left(\sum_{i=0}^{k-1} \psi_i(r)\mathbf{B}_x^{\top}(r)\mathbf{v}_i\right)^2\right]dr \\
&= \sum_{i,\ell} \mathbf{v}_i^{\top}\boldsymbol{\Omega}_{xx}\mathbf{v}_{\ell}\int_0^1 r\psi_i(r)\psi_{\ell}(r)dr.
\end{aligned}$$

Let $\mathbf{G}_{(k)}$ be the $k \times k$ matrix with (i,ℓ)th element equal to $\int_0^1 r\psi_i(r)\psi_{\ell}(r)dr$. Cauchy-Schwartz inequality gives

$$\left|\int_0^1 r\psi_i(r)\psi_{\ell}(r)dr\right| \leq \left(\int_0^1 r^2\psi_i^2(r)dr\right)^{1/2}\left(\int_0^1 r^2\psi_{\ell}^2(r)dr\right)^{1/2} \leq \int_0^1 \psi_i^2(r)dr = 1,$$

which follows from the fact that $\{\psi_i(r)\}_i$ is orthonormal. This implies that $\left\|\mathbf{G}_{(k)}\right\|_{sp} \leq 1$.

Substituting this above yields

$$\mathbb{E}\left[\int_0^1 g_{(k)}^2(r)dr\right] = \sum_{i,\ell} \mathbf{v}_i^\top \mathbf{\Omega}_{xx}\mathbf{v}_\ell \int_0^1 r\psi_i(r)\psi_\ell(r)dr$$
$$\leq \left\|\mathbf{G}_{(k)}\right\|_{sp} \sum_{i=0}^{k-1} \left\|\mathbf{\Omega}_{xx}^{1/2}\mathbf{v}_i\right\|^2 \leq \left\|\mathbf{\Omega}_{xx}^{1/2}\right\|_{sp}^2 \sum_{i=0}^{k-1} \left\|\mathbf{v}_i\right\|^2 < \infty,$$

which implies that $q_{(k)} = O_p(1)$. This completes the proof. □

Lemma C.3. Under Assumptions 1-3, we have, as $n \to \infty$

$$\frac{1}{n^2}\sum_{t=1}^n \left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\mathbf{X}_t\mathbf{X}_t^{\top(k)} = \int_0^1 \left(\mathbf{A}(r) - \mathbf{A}_k(r)\right)\left(\mathbf{B}_X(r)\right)\left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r)\right)^\top dr + O_p\left(\frac{1}{n^a k^{q-1/2}}\right),$$

where $a = \frac{p-2}{2p}$. Furthermore, $\left\|\int_0^1 \left(\mathbf{A}(r) - \mathbf{A}_k(r)\right)\left(\mathbf{B}_X(r)\right)\left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r)\right)^\top dr\right\|_{sp} = O_p\left(\frac{1}{k^q}\right)$.

Proof. Note that

$$\begin{aligned}
&\frac{1}{n^2}\sum_{t=1}^n \left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\mathbf{X}_t\mathbf{X}_t^{\top(k)} \\
&= \frac{1}{n^2}\sum_{t=1}^n \left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\left(\sqrt{n}\mathbf{V}_{n,t} + \tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0\right) \\
&\qquad\qquad \times \left(\boldsymbol{\psi}_{(k)}(t/n) \otimes \left(\sqrt{n}\mathbf{V}_{n,t} + \tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0\right)\right)^\top \\
&= \frac{1}{n^2}\sum_{t=1}^n \left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\left(\sqrt{n}\mathbf{V}_{n,t}\right)\left(\boldsymbol{\psi}_{(k)}(t/n) \otimes \left(\sqrt{n}\mathbf{V}_{n,t}\right)\right)^\top \\
&\quad + \frac{1}{n^2}\sum_{t=1}^n \left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\left(\sqrt{n}\mathbf{V}_{n,t}\right)\left(\boldsymbol{\psi}_{(k)}(t/n) \otimes \left(\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0\right)\right)^\top \\
&\quad + \frac{1}{n^2}\sum_{t=1}^n \left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\left(\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0\right)\left(\boldsymbol{\psi}_{(k)}(t/n) \otimes \sqrt{n}\mathbf{V}_{n,t}\right)^\top \\
&\quad + \frac{1}{n^2}\sum_{t=1}^n \left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\left(\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0\right)\left(\boldsymbol{\psi}_{(k)}(t/n) \otimes \left(\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0\right)\right)^\top \\
&:= \mathbf{\Delta}_n(1) + \mathbf{\Delta}_n(2) + \mathbf{\Delta}_n(3) + \mathbf{\Delta}_n(4),
\end{aligned}$$

so that

$$\left\| \frac{1}{n^2} \sum_{t=1}^{n} \left(\mathbf{A}_t - \mathbf{A}_k(t/n) \right) \mathbf{X}_t \mathbf{X}_t^{\top(k)} - \mathbf{\Delta}_n(1) \right\| \leq \left\| \mathbf{\Delta}_n(2) \right\| + \left\| \mathbf{\Delta}_n(3) \right\| + \left\| \mathbf{\Delta}_n(4) \right\|.$$

We first show that $\|\mathbf{\Delta}_n(2)\| = O_p\left(\frac{1}{k^{q-1/2}n^a}\right)$, $\|\mathbf{\Delta}_n(3)\| = O_p\left(\frac{1}{k^{q-1/2}n^a}\right)$, and $\|\mathbf{\Delta}_n(4)\| = O_p\left(\frac{1}{n^{2a}k^{q-1/2}}\right)$. This would imply that

$$\frac{1}{n^2} \sum_{t=1}^{n} \left(\mathbf{A}_t - \mathbf{A}_k(t/n) \right) \mathbf{X}_t \mathbf{X}_t^{\top(k)} = \mathbf{\Delta}_n(1) + O_p\left(\frac{1}{k^{q-1/2} n^a} \right). \tag{C.8}$$

Let us start with $\mathbf{\Delta}_n(2)$. We note that

$$\begin{aligned}
\|\mathbf{\Delta}_n(2)\| &= \left\| \frac{1}{n} \sum_{t=1}^{n} \left(\mathbf{A}_t - \mathbf{A}_k(t/n) \right) \left(\mathbf{V}_{n,t} \right) \left(\boldsymbol{\psi}_{(k)}(t/n) \otimes \frac{1}{\sqrt{n}} \left(\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0 \right) \right)^{\top} \right\| \\
&\leq \max_{1 \leq t \leq n} \left\| \frac{1}{\sqrt{n}} \left(\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0 \right) \right\| \frac{1}{n} \sum_{t=1}^{n} \left\| \left(\mathbf{A}_t - \mathbf{A}_k(t/n) \right) \left(\mathbf{V}_{n,t} \right) \right\| \left\| \boldsymbol{\psi}_{(k)}(t/n) \right\|.
\end{aligned}$$

By Lemma C.6 we have $\max_{1 \leq t \leq n} \left\| \frac{1}{\sqrt{n}} \left(\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0 \right) \right\| = O_p\left(\frac{1}{n^a}\right)$. In addition, since

$$\begin{aligned}
&\frac{1}{n} \sum_{t=1}^{n} \left\| \left(\mathbf{A}_t - \mathbf{A}_k(t/n) \right) \mathbf{V}_{n,t} \right\| \left\| \boldsymbol{\psi}_{(k)}(t/n) \right\| \\
&= \int_0^1 \left\| \left(\mathbf{A}(\lfloor nr \rfloor/n) - \mathbf{A}_k(\lfloor nr \rfloor/n) \right) \mathbf{V}_{n,\lfloor nr \rfloor} \right\| \left\| \boldsymbol{\psi}_{(k)}(\lfloor nr \rfloor/n) \right\| dr + O_p(1/n) \\
&= O_p\left(\frac{1}{k^{q-1/2}} \right) + o_p(1),
\end{aligned}$$

which follows from Cauchy-Schwarz inequality and (11):

$$\begin{aligned}
&\int_0^1 \left\| \left(\mathbf{A}(r) - \mathbf{A}_k(r) \right) \mathbf{V}_{n,\lfloor nr \rfloor} \right\| \left\| \boldsymbol{\psi}_{(k)}(r) \right\| dr \\
&\leq \left(\int_0^1 \left\| \mathbf{A}(r) - \mathbf{A}_k(r) \right\|^2 dr \right)^{1/2} \left(\int_0^1 \left\| \mathbf{V}_{n,\lfloor nr \rfloor} \right\|^2 dr \right)^{1/2} \left(\int_0^1 \left\| \boldsymbol{\psi}_{(k)}(r) \right\|^2 dr \right)^{1/2} \\
&\leq \left(\int_0^1 \left\| \mathbf{A}(r) - \mathbf{A}_k(r) \right\|^2 dr \right)^{1/2} \left(\int_0^1 \left\| \mathbf{V}_{n,\lfloor nr \rfloor} \right\|^2 dr \right)^{1/2} \left(\int_0^1 \left\| \boldsymbol{\psi}_{(k)}(r) \right\|^2 dr \right)^{1/2} \\
&= O_p\left(\frac{1}{k^{q-1/2}} \right).
\end{aligned}$$

We thus conclude that $\|\mathbf{\Delta}_n(2)\| = O_p\left(\frac{1}{k^{q-1/2}n^a}\right)$. Following similar argument, we can show that $\|\mathbf{\Delta}_n(3)\| = O_p\left(\frac{1}{k^{q-1/2}n^a}\right)$. For $\mathbf{\Delta}_n(4)$, we have

$$\begin{aligned}
&\|\mathbf{\Delta}_n(4)\| \\
=& \left\| \frac{1}{n}\sum_{t=1}^{n} \left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\left(\frac{1}{\sqrt{n}}\left(\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0\right)\right)\left(\boldsymbol{\psi}_{(k)}(t/n) \otimes \frac{1}{\sqrt{n}}\left(\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0\right)\right)^\top \right\| \\
\leq& \max_{1\leq t\leq n} \left\|\frac{1}{\sqrt{n}}\left(\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0\right)\right\| \max_{1\leq t\leq n} \left\|\frac{1}{\sqrt{n}}\left(\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0\right)\right\| \frac{1}{n}\sum_{t=1}^{n} \|\mathbf{A}_t - \mathbf{A}_k(t/n)\| \left\|\boldsymbol{\psi}_{(k)}(t/n)\right\| \\
=& O_p\left(\frac{1}{n^a}\right) O_p\left(\frac{1}{n^a}\right) O\left(\frac{1}{k^{q-1/2}}\right) = O_p\left(\frac{1}{n^{2a}k^{q-1/2}}\right) = o_p(1).
\end{aligned}$$

Finally, we consider $\mathbf{\Delta}_n(1)$. We have

$$\begin{aligned}
&\mathbf{\Delta}_n(1) \\
=& \frac{1}{n}\sum_{t=1}^{n} \left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right) \mathbf{V}_{n,t} \left(\boldsymbol{\psi}_{(k)}(t/n) \otimes \mathbf{V}_{n,t}\right)^\top \\
=& \int_0^1 \left(\mathbf{A}(\lfloor nr\rfloor/n) - \mathbf{A}_k(\lfloor nr\rfloor/n)\right)\left(\mathbf{V}_{n,\lfloor nr\rfloor}\right)\left(\boldsymbol{\psi}_{(k)}(\lfloor nr\rfloor/n) \otimes \mathbf{V}_{n,\lfloor nr\rfloor}\right)^\top dr + O_p(1/n) \\
=& \int_0^1 \left(\mathbf{A}(r) - \mathbf{A}_k(r)\right)\left(\mathbf{V}_{n,\lfloor nr\rfloor}\right)\left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{V}_{n,\lfloor nr\rfloor}\right)^\top dr + o_p(1),
\end{aligned}$$

where the second inequality follows from a Riemann–sum approximation of integral, the final equality follows from the continuity of $\mathbf{A}(\cdot)$ and the fact that $\left\|\boldsymbol{\psi}_{(k)}(\lfloor nr\rfloor/n) - \boldsymbol{\psi}_{(k)}(r)\right\| = O\left(\frac{k^{3/2}}{n}\right) = o(1)$ for all $r \in [0,1]$. This establishes (C.8). Finally, by Cauchy–Schwarz inequality, $\int_0^1 \left\|\boldsymbol{\psi}_{(k)}(r)\right\|^2 dr = kO(1)$, and $\sup_{r\in[0,1]} \left\|\mathbf{V}_{n,\lfloor nr\rfloor} - \mathbf{B}_X(r)\right\| = o_p(1)$, we could show that $\left\|\mathbf{R}_{\mathbf{\Delta}_n(1)}\right\| = O_p\left(k^{-(q-1/2)}\right) = o_p(1)$, where

$$\mathbf{R}_{\mathbf{\Delta}_n(1)} = \int_0^1 \left(\mathbf{A}(r) - \mathbf{A}_k(r)\right)\left\{\left(\mathbf{V}_{n,\lfloor nr\rfloor}\right)\left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{V}_{n,\lfloor nr\rfloor}\right)^\top - \left(\mathbf{B}_X(r)\right)\left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_X(r)\right)^\top\right\} dr.$$

Therefore,

$$\begin{aligned}
&\frac{1}{n^2}\sum_{t=1}^{n}\left(\mathbf{A}_t-\mathbf{A}_k(t/n)\right)\mathbf{X}_t\mathbf{X}_t^{\top(k)} \\
&=\int_0^1\left(\mathbf{A}(r)-\mathbf{A}_k(r)\right)\left(\mathbf{B}_X(r)\right)\left(\boldsymbol{\psi}_{(k)}(r)\otimes\mathbf{B}_X(r)\right)^\top dr+O_p\left(\frac{1}{k^{q-1/2}n^a}\right).
\end{aligned}$$

Now we show that $\left\|\int_0^1\left(\mathbf{A}(r)-\mathbf{A}_k(r)\right)\left(\mathbf{B}_X(r)\right)\left(\boldsymbol{\psi}_{(k)}(r)\otimes\mathbf{B}_X(r)\right)^\top dr\right\|_{sp}=O_p\left(\frac{1}{k^q}\right)$. Let $\mathbf{u}\in\mathbb{R}^d$ and $\mathbf{v}=\left(\mathbf{v}_0^\top,\cdots,\mathbf{v}_{k-1}^\top\right)^\top\in\mathbb{R}^{K_1k}$ be unit vectors such that $\|\mathbf{u}\|=1$ and $\|\mathbf{v}\|=1$. Furthermore, define

$$z_{(k)}:=\mathbf{u}^\top\int_0^1\left(\mathbf{A}(r)-\mathbf{A}_k(r)\right)\left(\mathbf{B}_X(r)\right)\left(\boldsymbol{\psi}_{(k)}(r)\otimes\mathbf{B}_X(r)\right)^\top dr\mathbf{v},$$

and

$$\mathbf{H}_{(k)}(r):=\mathbf{A}(r)-\mathbf{A}_k(r),\ \ \mathbf{G}_{(k)}(r):=\mathbf{u}\mathbf{B}_X^\top(r)\left(\boldsymbol{\psi}_{(k)}(r)\otimes\mathbf{B}_X(r)\right)^\top\mathbf{v}.$$

We then have

$$\begin{aligned}
|z_{(k)}|&=\left|\int_0^1\operatorname{tr}\left(\mathbf{H}_{(k)}^\top(r)\mathbf{G}_{(k)}(r)\right)dr\right| \\
&\le\left(\int_0^1\left\|\mathbf{H}_{(k)}(r)\right\|^2dr\right)^{1/2}\left(\int_0^1\left\|\mathbf{G}_{(k)}(r)\right\|^2dr\right)^{1/2}.
\end{aligned}$$

By (11), we have $\left(\int_0^1\left\|\mathbf{H}_{(k)}(r)\right\|^2dr\right)^{1/2}=O\left(1/k^q\right)$. We also have

$$\begin{aligned}
\int_0^1\left\|\mathbf{G}_{(k)}(r)\right\|^2dr&\le\int_0^1\|\mathbf{B}_X(r)\|^4\left\|\sum_{j=0}^{k-1}\psi_j(r)\mathbf{v}_j\right\|^2dr \\
&\le\sup_{r\in[0,1]}\|\mathbf{B}_X(r)\|^4\int_0^1\left\|\sum_{j=0}^{k-1}\psi_j(r)\mathbf{v}_j\right\|^2dr.
\end{aligned}$$

Observe that

$$\int_0^1 \left\| \sum_{j=0}^{k-1} \psi_j(r)\mathbf{v}_j \right\|^2 dr = \int_0^1 \sum_{j,\ell} \psi_j(r)\psi_\ell(r)\mathbf{v}_j^\top \mathbf{v}_\ell dr$$
$$= \sum_{j=0}^{k-1} \|\mathbf{v}_j\|^2 = \|\mathbf{v}\|^2 = 1,$$

which follows from the orthonormality of $\{\psi_j(r)\}_j$. This implies that $\int_0^1 \left\|\mathbf{G}_{(k)}(r)\right\|^2 dr = O_p(1)$, since $\sup_{r\in[0,1]} \ \|\mathbf{B}_X(r)\|^4 = O_p(1)$. □

Lemma C.4. Under Assumptions 1-3 and (19), we have, as $n \to \infty$

(i)

$$\frac{1}{n}\sum_{t=1}^n \mathbf{Z}_t\mathbf{X}_t^{\top(k)} = \mathbf{\Gamma}_{zx}^{(k)} + o_p(1),$$

where

$$\mathbf{\Gamma}_{zx}^{(k)} := \int_0^1 d\mathbf{B}_Z(r)\left(\boldsymbol{\psi}_{(k)}(r)\otimes \mathbf{B}_X(r)\right)^\top + \boldsymbol{\ell}_{1(k)} \otimes (\mathbf{\Lambda}_{zx} + \mathbf{\Sigma}_{zx})$$

satisfies $\left\|\mathbf{\Gamma}_{zx}^{(k)}\right\|_{sp} = O_p(1)$. $\boldsymbol{\ell}_{1(k)}$ denotes an $k\times 1$ vector with the first element equal to one and all remaining elements equal to zero.

(ii)

$$\left\|\frac{1}{n}\sum_{t=1}^n \left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\mathbf{X}_t\mathbf{Z}_t^\top\right\| = O_p\left(\frac{\sqrt{n}}{k^q}\right)$$

Proof. We only show (ii), as (i) is identical to Lemma C.2. Again, by BN decomposition, we have

$$\begin{aligned}
&\frac{1}{n}\sum_{t=1}^n \left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\mathbf{X}_t\mathbf{Z}_t^\top \\
&= \frac{1}{n}\sum_{t=1}^n \left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\left(\sqrt{n}\mathbf{V}_{n,t}^x + \tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0\right)\left(\mathbf{\Phi}_z(1)\boldsymbol{\varepsilon}_t - \Delta\tilde{\mathbf{u}}_{zt}\right)^\top \\
&= \frac{1}{n}\sum_{t=1}^n \left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\left(\sqrt{n}\mathbf{V}_{n,t}^x\right)\left(\mathbf{\Phi}_z(1)\boldsymbol{\varepsilon}_t\right)^\top - \frac{1}{n}\sum_{t=1}^n \left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\left(\sqrt{n}\mathbf{V}_{n,t}^x\right)\left(\Delta\tilde{\mathbf{u}}_{zt}\right)^\top \\
&\quad + \frac{1}{n}\sum_{t=1}^n \left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\left(\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0\right)\left(\mathbf{\Phi}_z(1)\boldsymbol{\varepsilon}_t\right)^\top - \frac{1}{n}\sum_{t=1}^n \left(\mathbf{A}_t - \mathbf{A}_k(t/n)\right)\left(\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0\right)\left(\Delta\tilde{\mathbf{u}}_{zt}\right)^\top \\
&:= \mathbf{\Delta}_n^z(1) + \mathbf{\Delta}_n^z(2) + \mathbf{\Delta}_n^z(3) + \mathbf{\Delta}_n^z(4).
\end{aligned}$$

Let us start with $\boldsymbol{\Delta}_n^z(1)$. By F.C.L.T. and C.M.T., we have $\sup_{1\le t\le n} \left\|\mathbf{V}_{n,t}^x\right\| = O_p(1)$. In addition, $\frac{1}{n}\sum_{t=1}^n \left\|\mathbf{A}_t - \mathbf{A}_k(t/n)\right\|^2 = \int_0^1 \left\|\mathbf{A}(r) - \mathbf{A}_k(r)\right\|^2 dr + O(1/n) = O(1/k^q)$, which follows from a Riemann–sum approximation of integral and (11). Under Assumption 1, we have $\frac{1}{n}\sum_{t=1}^n \left\|\boldsymbol{\Phi}_z(1)\boldsymbol{\varepsilon}_t\right\|^2 = O_p(1)$. Combining all these yields

$$\begin{aligned}\left\|\boldsymbol{\Delta}_n^z(1)\right\| &\le \frac{1}{n}\sum_{t=1}^n \left\|\mathbf{A}_t - \mathbf{A}_k(t/n)\right\| \left\|\sqrt{n}\mathbf{V}_{n,t}^x\right\| \left\|\boldsymbol{\Phi}_z(1)\boldsymbol{\varepsilon}_t\right\| \\ &\le \sup_{1\le t\le n} \left\|\sqrt{n}\mathbf{V}_{n,t}^x\right\| \left(\frac{1}{n}\sum_{t=1}^n \left\|\mathbf{A}_t - \mathbf{A}_k(t/n)\right\|^2\right)^{1/2} \left(\frac{1}{n}\sum_{t=1}^n \left\|\boldsymbol{\Phi}_z(1)\boldsymbol{\varepsilon}_t\right\|^2\right)^{1/2} \\ &= O_p(\sqrt{n})O_p(1/k^q) = O_p\left(\frac{\sqrt{n}}{k^q}\right),\end{aligned}$$

where the second equality follows from Cauchy-Schwarz inequality. Proceeding with similar reasoning as above, we could show that

$$\begin{aligned}\left\|\boldsymbol{\Delta}_n^z(2)\right\| &\le \frac{1}{n}\sum_{t=1}^n \left\|\mathbf{A}_t - \mathbf{A}_k(t/n)\right\| \left\|\sqrt{n}\mathbf{V}_{n,t}^x\right\| \left\|\Delta\tilde{\mathbf{u}}_{zt}\right\| \\ &\le \sup_{1\le t\le n} \left\|\sqrt{n}\mathbf{V}_{n,t}^x\right\| \left(\frac{1}{n}\sum_{t=1}^n \left\|\mathbf{A}_t - \mathbf{A}_k(t/n)\right\|^2\right)^{1/2} \left(\frac{1}{n}\sum_{t=1}^n \left\|\Delta\tilde{\mathbf{u}}_{zt}\right\|^2\right)^{1/2} \\ &= O_p(\sqrt{n})O_p(1/k^q) = O_p\left(\frac{\sqrt{n}}{k^q}\right),\end{aligned}$$

and

$$\begin{aligned}\left\|\boldsymbol{\Delta}_n^z(3)\right\| &\le \frac{1}{n}\sum_{t=1}^n \left\|\mathbf{A}_t - \mathbf{A}_k(t/n)\right\| \left\|\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0\right\| \left\|\Delta\tilde{\mathbf{u}}_{zt}\right\| \\ &\le \max_{1\le t\le n} \left\|\frac{1}{\sqrt{n}}\left(\tilde{\mathbf{u}}_{x0} - \tilde{\mathbf{u}}_{xt} + \mathbf{X}_0\right)\right\| \left(\frac{1}{n}\sum_{t=1}^n \left\|\mathbf{A}_t - \mathbf{A}_k(t/n)\right\|^2\right)^{1/2} \left(\frac{1}{n}\sum_{t=1}^n \left\|\Delta\tilde{\mathbf{u}}_{zt}\right\|^2\right)^{1/2} \\ &= O_p(\sqrt{n})O_p(\sqrt{n}/k^q) = O_p\left(\frac{n^{\frac{1}{2}-a}}{k^q}\right) = o_p(1),\end{aligned}$$

where the final equality follows from Lemma C.6. The case for $\left\|\boldsymbol{\Delta}_n^z(4)\right\| = o_p(1)$ is also similar.

□

Lemma C.5. For the model specification (24), let $\mathbf{Z}_t = \left(\Delta\mathbf{X}_{t-K}^\top, \cdots, \Delta\mathbf{X}_{t+K}^\top\right)^\top$. Under Assumptions 1-3, we have, as $n \to \infty$

(i) $\left\| \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \widetilde{\mathbf{v}}_t \mathbf{Z}_t^\top \right\| = O_p\left(\sqrt{K/n}\right)$;

(ii) $\left\| \left(\frac{1}{n-2K} \sum_{t=K+1}^{n-K} \mathbf{Z}_t \mathbf{Z}_t^\top \right)^{-1} - \mathbf{\Sigma}_{zz}^{-1} \right\|_{sp} = O_p\left(K/\sqrt{n}\right)$;

(iii) $\left\| \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \mathbf{Z}_t \mathbf{X}_t^{(k)\top} \right\| = O_p\left(\sqrt{Kk}\right)$;

(iv)

$$\frac{1}{n} \sum_{t=1}^{n} \mathbf{v}_t \mathbf{X}_t^{\top(k)} = \int_0^1 d\mathbf{B}_v(r) \left(\left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_r(\mathbf{\Omega}_{xx}) \right)^\top \right)^\top + o_p(1),$$

where $\left\| \int_0^1 d\mathbf{B}_v(r) \left(\left(\boldsymbol{\psi}_{(k)}(r) \otimes \mathbf{B}_r(\mathbf{\Omega}_{xx}) \right)^\top \right)^\top \right\|_{sp} = O_p(1)$.

Proof. Proof of (i). Recall that $\widetilde{\mathbf{v}}_t = \mathbf{v}_t + \sum_{|j|>K} \mathbf{\Gamma}_j \mathbf{X}_{t-j}$. Then,

$$\frac{1}{n-2K} \sum_{t=K+1}^{n-K} \widetilde{\mathbf{v}}_t \mathbf{Z}_t^\top = \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \mathbf{v}_t \mathbf{Z}_t^\top + \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \left(\sum_{|j|>K} \mathbf{\Gamma}_j \mathbf{X}_{t-j} \right) \mathbf{Z}_t^\top.$$

By Lemma A5 in Saikkonen 1991, we have $\left\| \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \left(\sum_{|j|>K} \mathbf{\Gamma}_j \mathbf{X}_{t-j} \right) \mathbf{Z}_t^\top \right\| = o_p\left(\sqrt{K/n}\right)$. In addition, Lemma A6 in Saikkonen 1991 gives $\left\| \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \mathbf{v}_t \mathbf{Z}_t^\top \right\| = O_p\left(\sqrt{K/n}\right)$. (i) follows accordingly.

Proof of (ii). This is just Lemma A4 in Saikkonen 1991 so details are omitted.

Proof of (iii). Since K_1 is fixed and $\mathbf{X}_0 = O_p(1)$, for notation simplicity we shall prove for the scalar case $K_1 = 1$ and assume that $X_0 = 0$. Note that

$$\begin{aligned}
\frac{1}{n-2K} \sum_{t=K+1}^{n-K} \mathbf{Z}_t \mathbf{X}_t^{(k)\top} &= \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \mathbf{Z}_t \left(\boldsymbol{\psi}_{(k)}(t/n) \otimes X_t \right)^\top \\
&= \frac{1}{n-2K} \sum_{t=K+1}^{n-K} \mathbf{Z}_t X_t \boldsymbol{\psi}_{(k)}^\top(t/n),
\end{aligned}$$

which is of dimension $(2K+1) \times k$. Consider a generic element $\frac{1}{n-2K} \sum_{t=K+1}^{n-K} \psi_r(t/n) X_t u_{x,t+j}$, where $j = 1, 2, \cdots, 2K+1, r = 1, 2, \cdots, k$. Observe that

$$\begin{aligned}
\mathbb{E}\left(X_t u_{x,t+j} X_s u_{x,s+j}\right) &= \sum_{\ell_1=1}^{t}\sum_{\ell_2=1}^{s} \mathbb{E}\left(u_{x,\ell_1} u_{x,t+j} u_{x,\ell_2} u_{x,s+j}\right) \\
&= \sum_{\ell_1=1}^{t}\sum_{\ell_2=1}^{s} \kappa_x(t+j-\ell_1, \ell_2-\ell_1, s+j-\ell_1) + \sum_{\ell_1=1}^{t}\sum_{\ell_2=1}^{s} \gamma_x(t+j-\ell_1)\gamma_x(s+j-\ell_2) \\
&\quad + \sum_{\ell_1=1}^{t}\sum_{\ell_2=1}^{s} \gamma_x(\ell_2-\ell_1)\gamma_x(s-t) + \sum_{\ell_1=1}^{t}\sum_{\ell_2=1}^{s} \gamma_x(s+j-\ell_1)\gamma_x(t+j-\ell_2),
\end{aligned}$$

where $\kappa_x(m_1, m_2, m_3) := \text{cum}\left(u_{x,t}, u_{x,t+m_1}, u_{x,t+m_2}, u_{x,t+m_3}\right)$ is the fourth order cumulant of $u_{x,t}$ and $\gamma_x(j)$ is the jth autocovariance of $u_{x,t}$. Then,

$$\begin{aligned}
&\mathbb{E}\left(\frac{1}{n-2K}\sum_{t=K+1}^{n-K} \psi_r(t/n) X_t u_{x,t+j}\right)^2 \\
&= \frac{1}{(n-2K)^2}\sum_t\sum_s \psi_r(t/n)\psi_r(s/n)\mathbb{E}\left(X_t u_{x,t+j} X_s u_{x,s+j}\right) \\
&= \frac{1}{(n-2K)^2}\sum_t\sum_s \psi_r(t/n)\psi_r(s/n)\sum_{\ell_1}\sum_{\ell_2}\kappa_x(t+j-\ell_1, \ell_2-\ell_1, s+j-\ell_1) \\
&\quad + \frac{1}{(n-2K)^2}\sum_t\sum_s \psi_r(t/n)\psi_r(s/n)\sum_{\ell_1}\sum_{\ell_2}\gamma_x(t+j-\ell_1)\gamma_x(s+j-\ell_2) \\
&\quad + \frac{1}{(n-2K)^2}\sum_t\sum_s \psi_r(t/n)\psi_r(s/n)\sum_{\ell_1}\sum_{\ell_2}\gamma_x(\ell_2-\ell_1)\gamma_x(s-t) \\
&\quad + \frac{1}{(n-2K)^2}\sum_t\sum_s \psi_r(t/n)\psi_r(s/n)\sum_{\ell_1}\sum_{\ell_2}\gamma_x(s+j-\ell_1)\gamma_x(t+j-\ell_2).
\end{aligned}$$

Now consider the first term above. Note that $|\psi_r(t/n)| \leq \sqrt{2}$ uniformly over any r and t. We have

$$
\begin{aligned}
&\left|\frac{1}{(n-2K)^2}\sum_t\sum_s \psi_r(t/n)\psi_r(s/n)\sum_{\ell_1}\sum_{\ell_2}\kappa_x(t+j-\ell_1,\ell_2-\ell_1,s+j-\ell_1)\right| \\
&\leq \frac{O(1)}{(n-2K)^2}\sum_t\sum_s\sum_{\ell_1}\sum_{m_2\in\mathbb{Z}}\left|\kappa_x(t+j-\ell_1,m_2,s+j-\ell_1)\right| \\
&\leq \frac{O(1)}{(n-2K)^2}\sum_t\sum_s\sum_{m_1\in\mathbb{Z}}\sum_{m_2\in\mathbb{Z}}\left|\kappa_x(m_1,m_2,m_1+s-t)\right| \\
&\leq \frac{O(1)}{(n-2K)^2}\sum_{m_1\in\mathbb{Z}}\sum_{m_2\in\mathbb{Z}}\sum_{m_3\in\mathbb{Z}}\left|\kappa_x(m_1,m_2,m_3)\right| = O\left(\frac{1}{n-2K}\right),
\end{aligned}
$$

since Assumption 1 guarantees that $\sum_{m_1\in\mathbb{Z}}\sum_{m_2\in\mathbb{Z}}\sum_{m_3\in\mathbb{Z}}\left|\kappa_x(m_1,m_2,m_3)\right| < \infty$. Moving on to the second term:

$$
\begin{aligned}
&\left|\frac{1}{(n-2K)^2}\sum_t\sum_s \psi_r(t/n)\psi_r(s/n)\sum_{\ell_1}\sum_{\ell_2}\gamma_x(t+j-\ell_1)\gamma_x(s+j-\ell_2)\right| \\
&\leq \frac{O(1)}{(n-2K)^2}\left(\sum_t\sum_{\ell_1}|\gamma_x(t+j-\ell_1)|\right)\left(\sum_s\sum_{\ell_2}|\gamma_x(s+j-\ell_2)|\right) \\
&\leq \frac{O(1)}{(n-2K)^2}\left\{\sum_t\left(\sum_{h_1\in\mathbb{Z}}|\gamma_x(h_1)|\right)\right\}\left\{\sum_s\left(\sum_{h_2\in\mathbb{Z}}|\gamma_x(h_2)|\right)\right\} = O(1),
\end{aligned}
$$

since Assumption 1(ii) guarantees that $\sum_{h\in\mathbb{Z}}|\gamma_x(h)| < \infty$. Using similar arguments, we could show that the third and fourth terms above are also $O(1)$. This implies that $\frac{1}{n-2K}\sum_{t=K+1}^{n-K}\psi_r(t/n)X_t u_{x,t+j} = O_p(1)$, so that $\left\|\frac{1}{n-2K}\sum_{t=K+1}^{n-K}\mathbf{Z}_t\mathbf{X}_t^{(k)\top}\right\| = O_p\left(\sqrt{Kk}\right)$. This completes the proof of (iii).

Proof of (iv). This is again similar to Lemma C.2 so details are omitted.

□

Lemma C.6. $\max_{1\leq t\leq n}\|\tilde{\boldsymbol{u}}_{xt}\| = O_p\left(n^{1/p}\right)$.

Proof. For the linear process $\mathbf{u}_{xt}$ defined in (3), we have the following Beveridge-Nelson (BN) decomposition

$$
\mathbf{u}_{xt} = \boldsymbol{\Phi}_x(1)\boldsymbol{\varepsilon}_t - \Delta\tilde{\mathbf{u}}_{xt},
$$

where $\boldsymbol{\Phi}_x(1) = \sum_{j=0}^{\infty}\boldsymbol{\Phi}_{xj}$, $\tilde{\mathbf{u}}_{xt} = \sum_{j=0}^{\infty}\widetilde{\boldsymbol{\Phi}}_{xj}\boldsymbol{\varepsilon}_{t-j}$, and $\widetilde{\boldsymbol{\Phi}}_{xj} = \sum_{k=j+1}^{\infty}\boldsymbol{\Phi}_{xk}$. By Minkowski inequality and the definition of L_p norm, we have

$$\|\tilde{\boldsymbol{u}}_{xt}\|_p = \left\|\sum_{j=0}^{\infty} \widetilde{\boldsymbol{\Phi}}_{xj}\boldsymbol{\varepsilon}_{t-j}\right\|_p \leq \sum_{j=0}^{\infty} \left\|\widetilde{\boldsymbol{\Phi}}_{xj}\boldsymbol{\varepsilon}_{t-j}\right\|_p = \sum_{j=0}^{\infty} \left(\mathbb{E}\left\|\widetilde{\boldsymbol{\Phi}}_{xj}\boldsymbol{\varepsilon}_{t-j}\right\|^p\right)^{1/p} = \|\boldsymbol{\varepsilon}_1\|_p \sum_{j=0}^{\infty} \left\|\widetilde{\boldsymbol{\Phi}}_{xj}\right\| < \infty,$$

where $\sum_{j=0}^{\infty}\left\|\widetilde{\boldsymbol{\Phi}}_{xj}\right\| < \infty$ follows from Lemma 2.1 in Phillips and Solo 1992. Then, by Markov inequality, we have, for any $M > 0$,

$$\mathbb{P}\left(n^{-1/p}\max_{1\leq t\leq n}|\tilde{\mathbf{u}}_{xt}| > M\right) \leq \sum_{t=1}^{n}\mathbb{P}\left(n^{-1/p}|\tilde{\mathbf{u}}_{xt}| > M\right) \leq \frac{n\mathbb{E}\,|\tilde{\mathbf{u}}_{xt}|^p}{(n^{1/p}M)^p} < \infty,$$

which implies that $\max_{1\leq t\leq n}\ \|\tilde{\boldsymbol{u}}_{xt}\| = O_p\left(n^{1/p}\right)$. □

D — Chebyshev time polynomials

Recall that Chebyshev time polynomials are defined by

$$\psi_0(t/n) = 1, \quad \psi_i(t/n) = \sqrt{2}\cos\left(i\pi(t/n)\right). \tag{D.1}$$

For a particular time coordinate, say $t = \lfloor nr \rfloor$ for $r \in [0,1]$, we have

$$\psi_i(\lfloor nr \rfloor/n) \sim \sqrt{2}\cos(i\pi r),$$

so that we can redefine the Chebyshev time polynomials for any $r \in [0,1]$:

$$\psi_0(r) = 1, \quad \psi_i(r) = \sqrt{2}\cos(i\pi r). \tag{D.2}$$

The Chebyshev polynomials form an orthonormal basis in $L^2[0,1] = \left\{a(r) : \int_0^1 a^2(r)dr < \infty\right\}$, in which the inner product is given by $\langle a_1, a_2\rangle = \int_0^1 a_1(r)a_2(r)dr$ and the induced norm $\|a\| =$

$\langle a, a \rangle$. To see this, notice that, for $i \neq j$, we have

$$\begin{aligned}
\langle \psi_i, \psi_j \rangle &= \int_0^1 \psi_i(r)\psi_j(r)dr \\
&= 2\int_0^1 \cos(i\pi r)\cos(j\pi r)dr \\
&= \int_0^1 \left[\cos\left((i+j)\pi r\right) + \cos\left((i-j)\pi r\right)\right] dr \\
&= \left[\frac{\sin\left((i+j)\pi r\right)}{(i+j)\pi} + \frac{\sin\left((i-j)\pi r\right)}{(i-j)\pi}\right]\Bigg|_0^1 = 0.
\end{aligned}$$

If $i = j = 0$, $\|\psi_i\| = 1$ holds trivially. For $i = j \neq 0$, we have

$$\begin{aligned}
\|\psi_i\| &= 2\int_0^1 \cos^2(i\pi r)dr \\
&= \int_0^1 \left[1 + \cos(2i\pi r)\right] dr \\
&= 1 + \left[\frac{1}{2i\pi}\sin(2i\pi r)\right]\Bigg|_0^1 = 1.
\end{aligned}$$

The Chebyshev series expansion for a generic function $a(r) : [0, 1] \to \mathbb{R}$ is defined as

$$a(r) \sim \sum_{i=0}^{\infty} b_i \psi_i(r). \tag{D.3}$$

Now taking inner product on (D.3) gives

$$\langle a, \psi_k \rangle = \sum_{i=0}^{\infty} b_i \langle \psi_i, \psi_k \rangle = b_k \langle \psi_k, \psi_k \rangle.$$

Since $\langle \psi_k, \psi_k \rangle = 2\int_0^\pi \cos^2 i\theta d\theta = \int_0^\pi (1 + \cos 2i\theta) = \pi + \frac{1}{2i\sin 2i\theta}\Big|_0^\pi = \pi$, we have

$$b_k = \frac{1}{\pi}\int_0^1 \frac{a(r)\psi_k(r)}{\sqrt{1-r^2}}dr,$$

for $k = 0, 1, 2, \cdots$.

We write $\left(S^{\intercal}_{m}a\right)(r)$ as the partial sum of the Chebyshev series of degree m:

$$\left(S^{\intercal}_{m}a\right)(r) = \sum_{i=0}^{m} b_i \psi_i(r).$$

Then, by Theorem 5.2 in Mason and Handscomb 2002, we know that the Chebyshev series expansion converges in $\mathcal{L}_2$ in the sense that

$$\int_0^1 \frac{\left[a(r) - \left(S^{\intercal}_{m}a\right)(r)\right]^2}{\sqrt{1-r^2}} dr \to 0, \tag{D.4}$$

as $m \to \infty$. However, (D.4) is not enough for our purpose, as we need the explicit convergence rate. This is provided in the following theorem.

Theorem D.1. If $a(r) : [0,1] \to \mathbb{R}$ has $q+1$ continuous derivatives, then $\left|\left(S^{\intercal}_{m}a\right)(r) - a(r)\right| = O\left(\frac{1}{m^q}\right)$ for any $r \in [0,1]$.

Proof. We follow closely the steps as the proofs of Theorem 5.14 in Mason and Handscomb 2002. We first introduce the operator L_m, which is defined by the relationship

$$L_m a := \left(S^{\intercal}_{m}a\right)(r) - a(r).$$

Let Π_n be the linear space of polynomials of degree n. For any p_q in Π_q, based on the definition of the operator L_m and $S^{\intercal}_{m}$, we have $S^{\intercal}_{m} p_q \equiv p_q$ so that $L_m p_q = S^{\intercal}_{m} p_q - p_q = 0$.

Using Peano' s theorem (Davis 1975), we obtain

$$\left(S^{\intercal}_{m}a\right)(r) - a(r) = \int_0^1 a^{(q+1)}(r) K_m(r,t) dt,$$

where $K_m(r,t) = \frac{1}{q!}\left\{S_m(r-t)^q_+ - (r-t)^q_+\right\}$, $(r-t)^q_+ = (r-t)^q$ if $r \geq t$ and zero otherwise. We note that

$$S_m(r-t)^q_+ = \sum_{i=0}^{m} b_{iq} \psi_i(r),$$

where $b_{iq} = \frac{1}{\pi}\int_t^1 \frac{(r-t)^m \psi_i(r)}{\sqrt{1-r^2}} dr$. By change of variables $r = \cos\theta$, $t = \cos\phi$, we obtain

$$b_{iq} = \frac{1}{\pi}\int_0^{\phi} (\cos\theta - \cos\phi)^q \cos i\theta \, d\theta.$$

We then show that $b_{iq} = O\left(\frac{1}{i^{q+1}}\right)$. To see this, notice that applying integration by parts once gives

$$\begin{aligned}
\left|\int_0^\phi (\cos\theta - \cos\phi)^q \cos i\theta \, d\theta\right| =& \frac{1}{i}\left|\int_0^\phi (\cos\theta - \cos\phi)^q \, d\sin i\theta\right| \\
=& \frac{1}{i}\left|\int_0^\phi \sin i\theta d\left(\cos\theta - \cos\phi\right)^q\right| \\
=& \frac{q}{i}\left|\int_0^\phi (\cos\theta - \cos\phi)^{q-1} \sin\theta \sin i\theta d\theta\right| \\
=& \frac{q}{i}\left|\int_0^\phi (\cos\theta - \cos\phi)^{q-1} \sin\theta \sin i\theta d\theta\right| \\
=& \frac{q}{2i}\left|\int_0^\phi (\cos\theta - \cos\phi)^{q-1} \left[\cos\left((i-1)\theta\right) - \cos\left((i+1)\theta\right)\right] d\theta\right|
\end{aligned} \tag{D.5}$$

$$\leq \frac{q}{2i}\left|(\cos\theta - \cos\phi)^{q-1}\right| \left|\cos\left((i-1)\theta\right) - \cos\left((i+1)\theta\right)\right| \leq \frac{O(1)}{i}.$$

If we continue from (D.5), we see that there are two terms:

$$\frac{q}{2i}\int_0^\phi (\cos\theta - \cos\phi)^{q-1} \cos\left((i-1)\theta\right) d\theta, \quad \frac{q}{2i}\int_0^\phi (\cos\theta - \cos\phi)^{q-1} \cos\left((i+1)\theta\right) d\theta.$$

Applying integration by parts on $\int_0^\phi (\cos\theta - \cos\phi)^{q-1} \cos\left((i-1)\theta\right) d\theta$ gives

$$\begin{aligned}
\left|\int_0^\phi (\cos\theta - \cos\phi)^{q-1} \cos\left((i-1)\theta\right) d\theta\right| &= \frac{(q-1)}{(i-1)}\left|\int_0^\phi (\cos\theta - \cos\phi)^{q-2} \sin\theta \sin\left((i-1)\theta\right) d\theta\right| \\
&= \frac{(q-1)}{2(i-1)}\left|\int_0^\phi (\cos\theta - \cos\phi)^{q-2} \left[\cos\left((i-2)\theta\right) - \cos\left(i\theta\right)\right] d\theta\right| \\
&\leq \frac{O(1)}{i-1}.
\end{aligned}$$

Applying integration by parts on $\int_0^\phi (\cos\theta - \cos\phi)^{q-1} \cos\left((i+1)\theta\right) d\theta$ gives

$$\begin{aligned}
\left|\int_0^\phi (\cos\theta - \cos\phi)^{q-1} \cos\left((i+1)\theta\right) d\theta\right| &= \frac{(q-1)}{(i+1)}\left|\int_0^\phi (\cos\theta - \cos\phi)^{q-2} \sin\theta \sin\left((i+1)\theta\right) d\theta\right| \\
&= \frac{(q-1)}{2(i+1)}\left|\int_0^\phi (\cos\theta - \cos\phi)^{q-2} \left[\cos\left(i\theta\right) - \cos\left((i+2))\theta\right)\right] d\theta\right| \\
&\leq \frac{O(1)}{i+1}.
\end{aligned}$$

This indicates that $\int_0^\phi (\cos\theta - \cos\phi)^{q-1} \cos((i+1)\theta)\, d\theta$ is just smaller order term. We continue from (D.5) to obtain

$$\begin{aligned}
\left| \int_0^\phi (\cos\theta - \cos\phi)^q \cos i\theta \, d\theta \right| \leq & \frac{q}{2i} \left| \int_0^\phi (\cos\theta - \cos\phi)^{q-1} \left[\cos((i-1)\theta) - \cos((i+1)\theta)\right] d\theta \right| \\
\leq & \frac{q}{2i} \left| \int_0^\phi (\cos\theta - \cos\phi)^{q-1} \cos((i-1)\theta)\, d\theta \right| \\
\leq & \frac{O(1)}{i(i-1)}
\end{aligned}$$

As $a(r)$ is assumed to have $q+1$ continuous derivative, we can apply integration by parts q times, so that

$$\left| \int_0^\phi (\cos\theta - \cos\phi)^q \cos i\theta \, d\theta \right| \leq \frac{O(1)}{i(i-1)\cdots(i-q)} = O\left(\frac{1}{i^{q+1}}\right),$$

which implies that $b_{iq} = O\left(\frac{1}{i^{q+1}}\right)$.

Finally, since

$$\begin{aligned}
\left|S_m (r-t)_+^q - (r-t)_+^q\right| = \left| \sum_{i=m+1}^{\infty} b_{iq} \psi_i(r) \right| \leq \sum_{i=m+1}^{\infty} |b_{iq}| = \frac{1}{m^q} \sum_{i=m+1}^{\infty} \frac{1}{m} \frac{1}{\left(\frac{i}{m}\right)^{q+1}} \\
= \frac{1}{m^q} \int_1^\infty \frac{1}{x^{q+1}} dx + o(1) = O\left(\frac{1}{m^q}\right),
\end{aligned}$$

so that

$$\left|\left(S_m^\intercal a\right)(r) - a(r)\right| \leq \left|a^{(q+1)}(r)\right| \left|K_m(r,t)\right| = O\left(\frac{1}{m^q}\right).$$

□